FIT PM12 twelve-channel test campaign

FIT hardware acceptance bench

November 13, 2026

1 Scope and recorded coverage

Board PM12-K7-3A17A1B402212223. Campaign 5ef2e2b4-c04a-4fad-b86f-a23be621d7a7.

Recorded status: INCOMPLETE. Missing tests are NOT MEASURED. Complete selected-protocol coverage is not a calibrated whole-board release.

Channel	cfd-monitor	cfd-dacs	gate-phase	tdc	adc-dc	adc-lines	integrator
1	NM	NM	NM	NM	NM	NM	NM
2	NM	NM	NM	NM	NM	NM	NM
3	NM	NM	NM	NM	NM	NM	NM
4	NM	NM	NM	NM	NM	NM	NM
5	NM	NM	NM	NM	NM	NM	NM
6	NM	NM	NM	NM	NM	NM	NM
7	NM	NM	NM	NM	NM	NM	NM
8	NM	NM	NM	NM	NM	NM	NM
9	NM	PASS	NM	NM	NM	NM	NM
10	NM	PASS	NM	NM	NM	NM	NM
11	NM	PASS	NM	NM	NM	NM	NM
12	NM	PASS	NM	NM	PASS	PASS	PASS

NM = not measured; INC = incomplete; ERR = acquisition error. No missing series is invented.

1.1 Board-level tests

Test	Status
Boundary scan and CPLD–FPGA connections	NOT_MEASURED
MCU, CPU–FPGA communication and peripheral IDs	NOT_MEASURED
I2C sensors, currents and voltages	NOT_MEASURED
HDMI digital link	NOT_MEASURED
Ethernet/bridge diagnostic	NOT_MEASURED

2 Fitted batches and fixture provenance

Batch attempt	PM channel	CFD label	Declared connection	Fitting
ch09-12-attempt001	12	CFD01	slot 7, RJ45 0	Identity/fitting confirmed
ch09-12-attempt001	11	CFD02	slot 7, RJ45 1	Identity/fitting confirmed
ch09-12-attempt001	10	CFD03	slot 7, RJ45 2	Identity/fitting confirmed
ch09-12-attempt001	9	CFD04	slot 7, RJ45 3	Identity/fitting confirmed

Fixture labels CFD01–CFD04 are provisional bench labels. Factory identities and photos remain unknown until supplied; a label is not proof of module identity. Declared cable routes must be checked physically.

Raw inputs and selected result IDs are retained in inputs/. Repeated measurements are never silently substituted.

3 Retained attempts and explicit selection

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-b9126235	adc-dc	12	ch09-12-attempt001	PASS	yes
result-ee0613ea	adc-lines	12	ch09-12-attempt001	PASS	yes
result-217f5cb4	integrator	12	ch09-12-attempt001	PASS	yes
result-b99efe3a	cfd-dacs	9	ch09-12-attempt001	INCOMPLETE	no
result-ef50493e	cfd-dacs	10	ch09-12-attempt001	INCOMPLETE	no
result-c4800a66	cfd-dacs	11	ch09-12-attempt001	INCOMPLETE	no
result-c80f3b9e	cfd-dacs	12	ch09-12-attempt001	INCOMPLETE	no
result-9e64b43e	cfd-dacs	9	ch09-12-attempt001	PASS	yes
result-0fcaadd7	cfd-dacs	10	ch09-12-attempt001	PASS	yes
result-8b6b8aa7	cfd-dacs	11	ch09-12-attempt001	PASS	yes
result-b4c6c497	cfd-dacs	12	ch09-12-attempt001	PASS	yes

All attempts, including failures and rejected repeats, remain in the archived inputs. The coverage matrix uses only the explicitly selected attempt.

4 Selected measurement conditions and limits

Test	Physical channel	Confirmed stimulus connec tion
adc-dc	12	{”dac_channels”: [0, 1], ”stimulus_channel”: 12, ”usb_input”: ”J8”}
adc-lines	12	{”dac_channels”: [0, 1], ”stimulus_channel”: 12, ”usb_input”: ”J8”}
integrator	12	{”dac_channels”: [2, 3], ”stimulus_channel”: 12, ”usb_input”: ”J6”}
cfd-dacs	9	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monito r_channels”: [9, 10, 11, 12]}
cfd-dacs	10	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monito r_channels”: [9, 10, 11, 12]}
cfd-dacs	11	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monito r_channels”: [9, 10, 11, 12]}
cfd-dacs	12	{”selection”: ”one monitor mux enabled; all others inhibited”, ”shared_monito r_channels”: [9, 10, 11, 12]}

4.1 Common hardware configuration

Setting	Recorded value
clock_profile.adc_rate	2x reference
clock_profile.kasli_rate	3x reference
clock_profile.reference_mhz	40.0787
cpld_gate_mode	original INTA/INTB enabled, nomina l20MHz
kasli_build.ABI	CFD4
kasli_build.expected_bit_sha256	a71a08a258620b0e5a52c5d03aefd8c69040 c9d15e8de6391c484ac182b7173b
pm_build	c7c42d28

4.2 adc-dc

Setting	Recorded value
method	”direct-ADC-DC-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400

Setting	Recorded value
integrator_input_parked_V	[2.0480001, 1.02400005]

4.3 adc-lines

Setting	Recorded value
method	”adjacent-pair-states-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

4.4 cfd-dacs

Setting	Recorded value
method	”shared-monitor-SMU-BCD-v1”
instrument_identity	”Keysight Technologies,B2910BL,MY6332 0299,5.0.2429.8966”
sense	”Force HI/LO, local”
source_current_A	0
source_range_A	1e-06
voltage_range_V	20
nplc	1
settling_s	0.1
points	101
repeats	3
limits	{”repeat_std_V”: 0.001, ”reverse_step_ V”: 0.0002, ”span_relative_error”: 0.05}
post_scan_verification	”SMU OFF/HIZ, four monitor muxes inhibi ted, explicit TCA masks0”

4.5 integrator

Setting	Recorded value
method	”signed-integration-pair-v1”
dac_channels	[2, 3]
base_V	[1.024, 2.048]
coarse_offsets_V	[-0.09999999999999987, -0.080000000000 00007, -0.05999999999999983, -0.040000 000000000036, -0.019999999999999796, -0.010000000000000009, 0.0, 0.009999999999 999787, 0.02000000000000024, 0.040000 000000000036, 0.060000000000000275, 0.08000000000000007, 0.10000000000000009]
fine_method	”26 points,0.4mV step around coarse estima te”

Setting	Recorded value
return_method	”two candidate revisits and +/-5mV offset s;near-zero <=2codes; signed response >4codes”

Commanded DAC voltages, whole connected chain; not calibrated ADC-only INL/noise or electrical continuity proof.

Effective differential whole-chain balance; waveform-based integration pair, not an absolute gate timestamp or calibrated opamp bias.

Functional assembled-chain transfer check: span within5% of schematic scaling, reverse step <=0.2mV, repeat standard deviation <=1mV. Not traceable calibration or exhaustive16-bit INL/DNL. DAC-A unchanged; exercised B/C/D end at0000..

ADC total RMS includes the complete connected chain and periodic gate feedthrough. The four-phase-removed RMS removes the mean of each sample phase; it is a residual statistic, not an intrinsic ADC noise specification. FFT and time axes use the declared clock frequency, not an independent frequency measurement.

5 Measurements and common channel plots

Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	A	gain	-1784.205440671235	code/V
adc-dc	12	A	zero_mean	2162.0811339198435	code
adc-dc	12	A	zero_total_rms	56.22754588987761	code
adc-dc	12	A	zero_four_phase_removed_rms	0.5926371250728474	code
adc-dc	12	B	gain	-1784.2694869545635	code/V
adc-dc	12	B	zero_mean	2165.8475073313784	code
adc-dc	12	B	zero_total_rms	64.40894678040004	code
adc-dc	12	B	zero_four_phase_removed_rms	0.5625571866355403	code
adc-lines	12	A	adjacent_pair_states	44	state
adc-lines	12	A	visited_codes	3147	code
adc-lines	12	B	adjacent_pair_states	44	state
adc-lines	12	B	visited_codes	3184	code
integrator	12	A	balance_delta	0.006030008721686399	V
integrator	12	A	gain	-2530.411960665549	code/V
integrator	12	A	near_balance_change	0.13016491045236994	code
integrator	12	B	balance_delta	0.007424191555263186	V

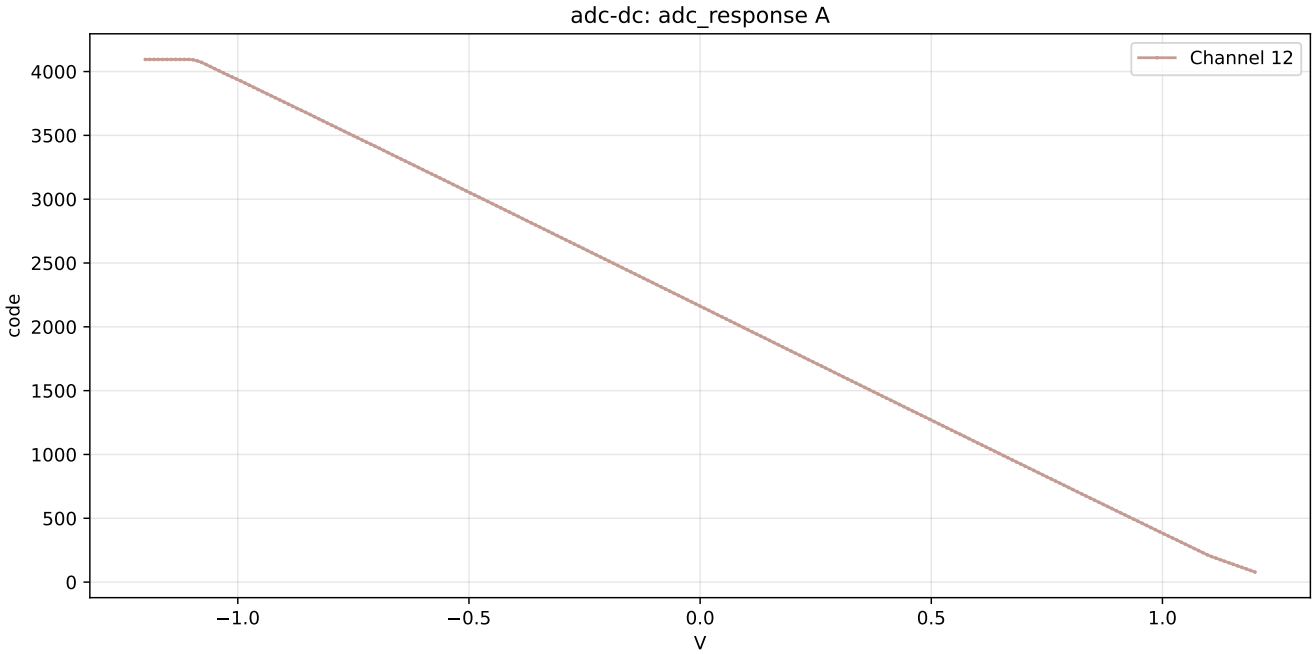
Test	Channel	Bank	Observable	Value	Unit
integrator	12	B	gain	-2448.007789 5617496	code/V
integrator	12	B	near_balance_change	0.111943 22217075807	code
cfd-dacs	9		dac_B_minimum_code_V	1.044726 6666666666	V
cfd-dacs	9		dac_B_maximum_code_V	1.004613 3333333334	V
cfd-dacs	9		dac_B_span_V	-0.040113 33333333322	V
cfd-dacs	9		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_B_maximum_repeat_std_V	7.023769 168570249e-05	V
cfd-dacs	9		dac_B_interior_gain_error_percent	-0.006716 857635891138	%
cfd-dacs	9		dac_C_minimum_code_V	1.034956 6666666667	V
cfd-dacs	9		dac_C_maximum_code_V	1.014396 6666666666	V
cfd-dacs	9		dac_C_span_V	-0.020560 000000000134	V
cfd-dacs	9		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_C_maximum_repeat_std_V	9.504384 952928754e-05	V
cfd-dacs	9		dac_C_interior_gain_error_percent	0.851897 1864797997	%
cfd-dacs	9		dac_D_minimum_code_V	0.0054	V
cfd-dacs	9		dac_D_maximum_code_V	1.06306	V
cfd-dacs	9		dac_D_span_V	1.057659 9999999998	V
cfd-dacs	9		dac_D_maximum_reverse_step_V	0	V

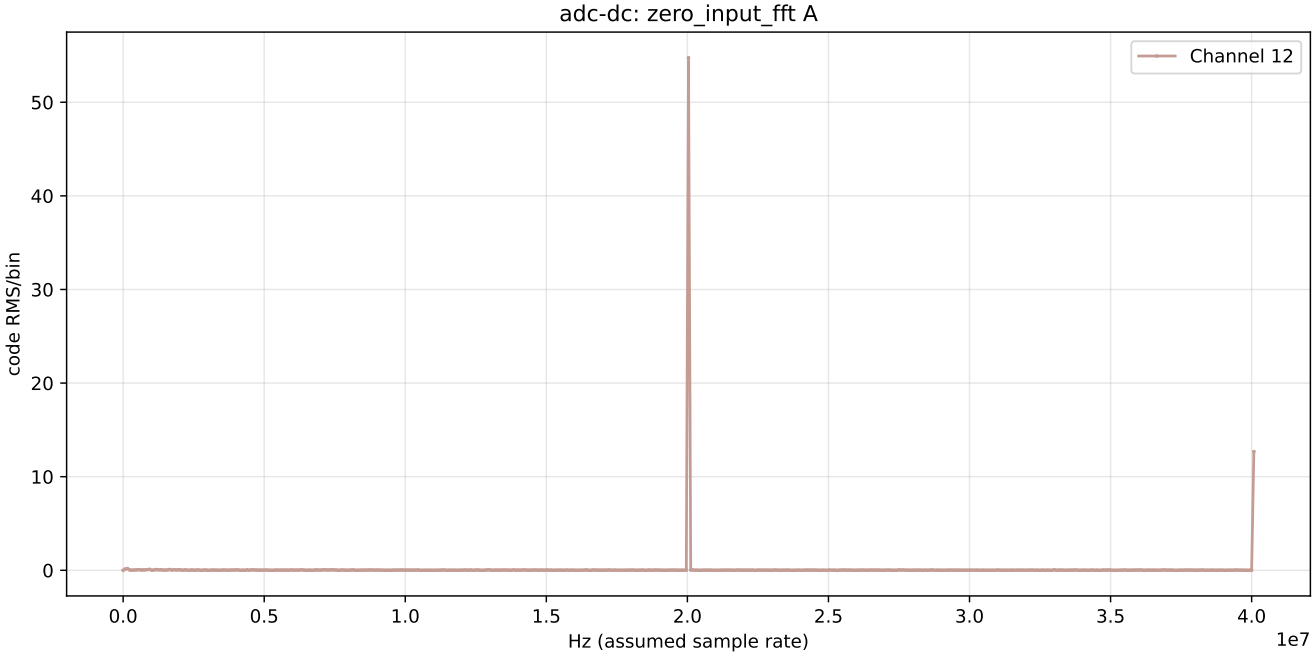
Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	9		dac_D_maximum_repeat_std_V	7.000000000000657e-05	V
cfd-dacs	9		dac_D_interior_gain_error_percent	-0.026880429186237897	%
cfd-dacs	10		dac_B_minimum_code_V	1.04437	V
cfd-dacs	10		dac_B_maximum_code_V	1.00438	V
cfd-dacs	10		dac_B_span_V	-0.039989999999999997	V
cfd-dacs	10		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_B_maximum_repeat_std_V	8.082903768656434e-05	V
cfd-dacs	10		dac_B_interior_gain_error_percent	-0.14222304801190244	%
cfd-dacs	10		dac_C_minimum_code_V	1.0346766666666667	V
cfd-dacs	10		dac_C_maximum_code_V	1.0141433333333334	V
cfd-dacs	10		dac_C_span_V	-0.020533333333292	V
cfd-dacs	10		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_C_maximum_repeat_std_V	9.539392001416794e-05	V
cfd-dacs	10		dac_C_interior_gain_error_percent	0.5828293358783654	%
cfd-dacs	10		dac_D_minimum_code_V	0.004546666666666667	V
cfd-dacs	10		dac_D_maximum_code_V	1.0692933333333334	V
cfd-dacs	10		dac_D_span_V	1.0647466666666667	V

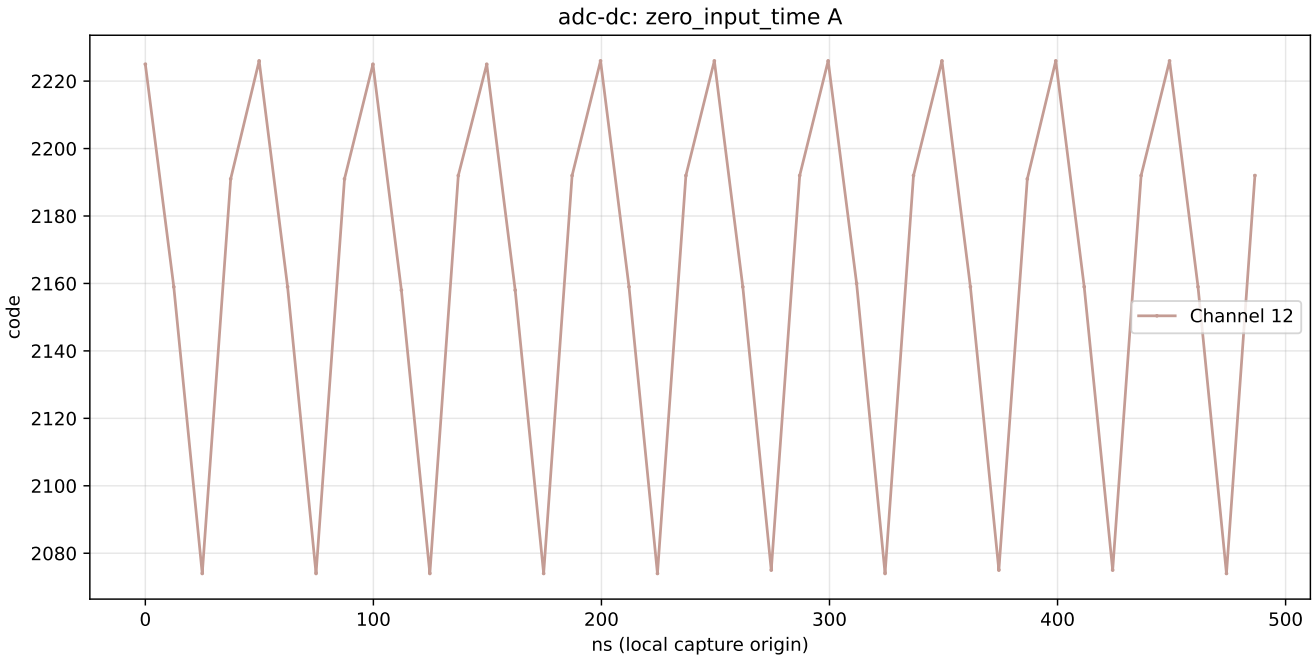
Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	10		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_D_maximum_repeat_std_V	7.000000000003828e-05	V
cfd-dacs	10		dac_D_interior_gain_error_percent	0.042824877925129456	%
cfd-dacs	11		dac_B_minimum_code_V	1.0446666666666666	V
cfd-dacs	11		dac_B_maximum_code_V	1.00461	V
cfd-dacs	11		dac_B_span_V	-0.040056666666666663	V
cfd-dacs	11		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_B_maximum_repeat_std_V	7.549834435276095e-05	V
cfd-dacs	11		dac_B_interior_gain_error_percent	-0.1743207749736131	%
cfd-dacs	11		dac_C_minimum_code_V	1.0349033333333333	V
cfd-dacs	11		dac_C_maximum_code_V	1.0143866666666665	V
cfd-dacs	11		dac_C_span_V	-0.020516666666674	V
cfd-dacs	11		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_C_maximum_repeat_std_V	7.234178138066471e-05	V
cfd-dacs	11		dac_C_interior_gain_error_percent	0.17720494086375105	%
cfd-dacs	11		dac_D_minimum_code_V	0.00485	V
cfd-dacs	11		dac_D_maximum_code_V	1.0636333333333334	V

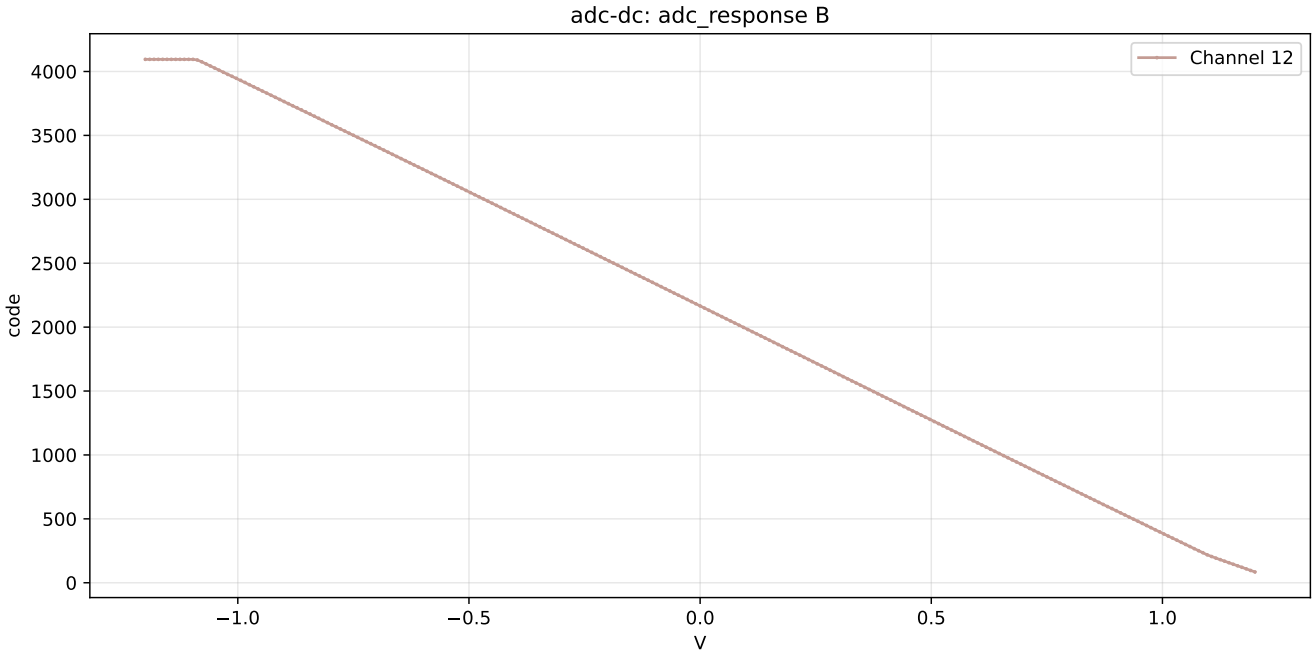
Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	11		dac_D_span_V	1.0587833333333334	V
cfd-dacs	11		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_D_maximum_repeat_std_V	6.110100926609052e-05	V
cfd-dacs	11		dac_D_interior_gain_error_percent	0.026777441441017125	%
cfd-dacs	12		dac_B_minimum_code_V	1.04463	V
cfd-dacs	12		dac_B_maximum_code_V	1.0044266666666666	V
cfd-dacs	12		dac_B_span_V	-0.04020333333333337	V
cfd-dacs	12		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_B_maximum_repeat_std_V	4.9328828623155544e-05	V
cfd-dacs	12		dac_B_interior_gain_error_percent	0.06510873936658168	%
cfd-dacs	12		dac_C_minimum_code_V	1.0348066666666667	V
cfd-dacs	12		dac_C_maximum_code_V	1.0144333333333333	V
cfd-dacs	12		dac_C_span_V	-0.020373333333333354	V
cfd-dacs	12		dac_C_maximum_reverse_step_V	3.3333333335772153e-06	V
cfd-dacs	12		dac_C_maximum_repeat_std_V	0.00011846237095938458	V
cfd-dacs	12		dac_C_interior_gain_error_percent	-1.395027937486315	%
cfd-dacs	12		dac_D_minimum_code_V	0.00498	V

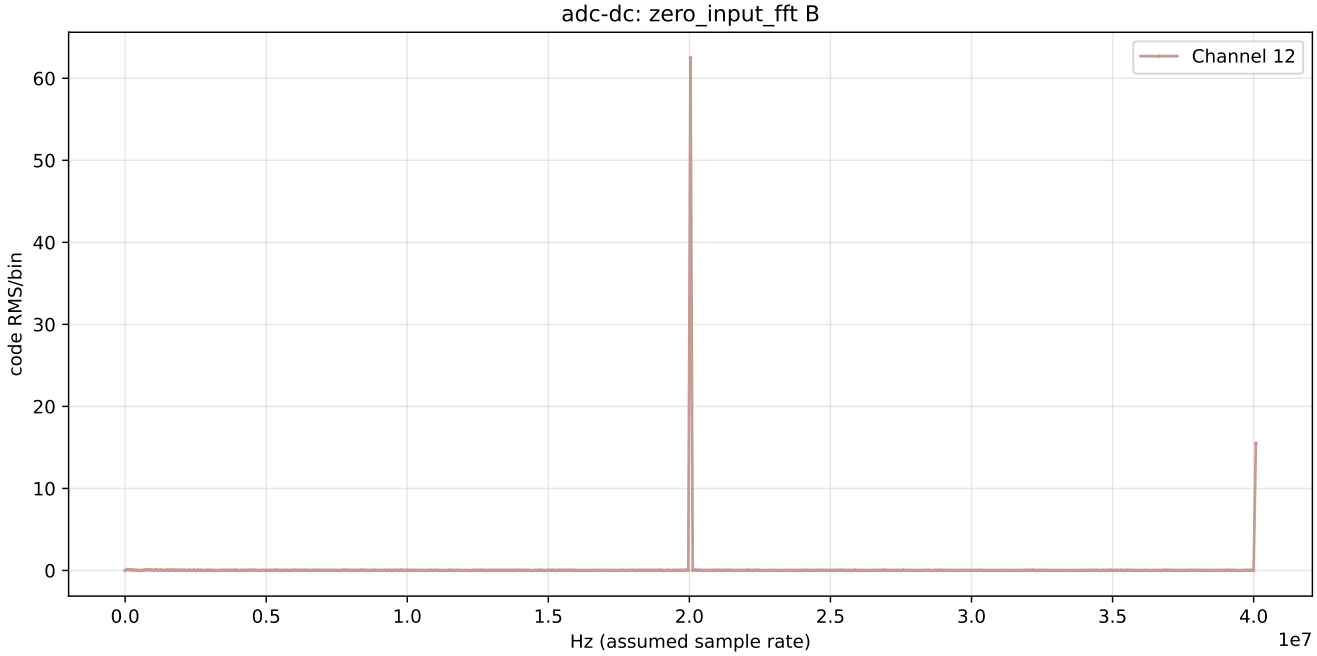
Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	12		dac_D_maximum_code_V	1.06321	V
cfd-dacs	12		dac_D_span_V	1.05823	V
cfd-dacs	12		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_D_maximum_repeat_std_V	8.020806 27701036e-05	V
cfd-dacs	12		dac_D_interior_gain_error_percent	-0.037869 6257134159	%

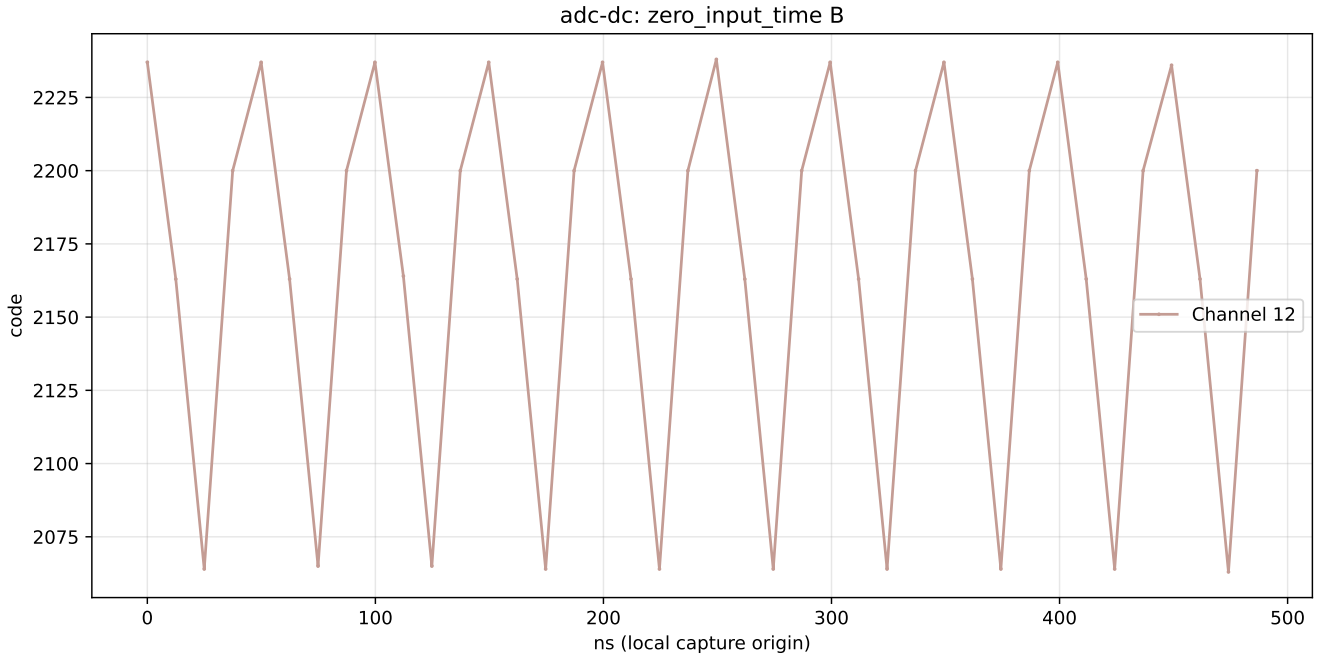


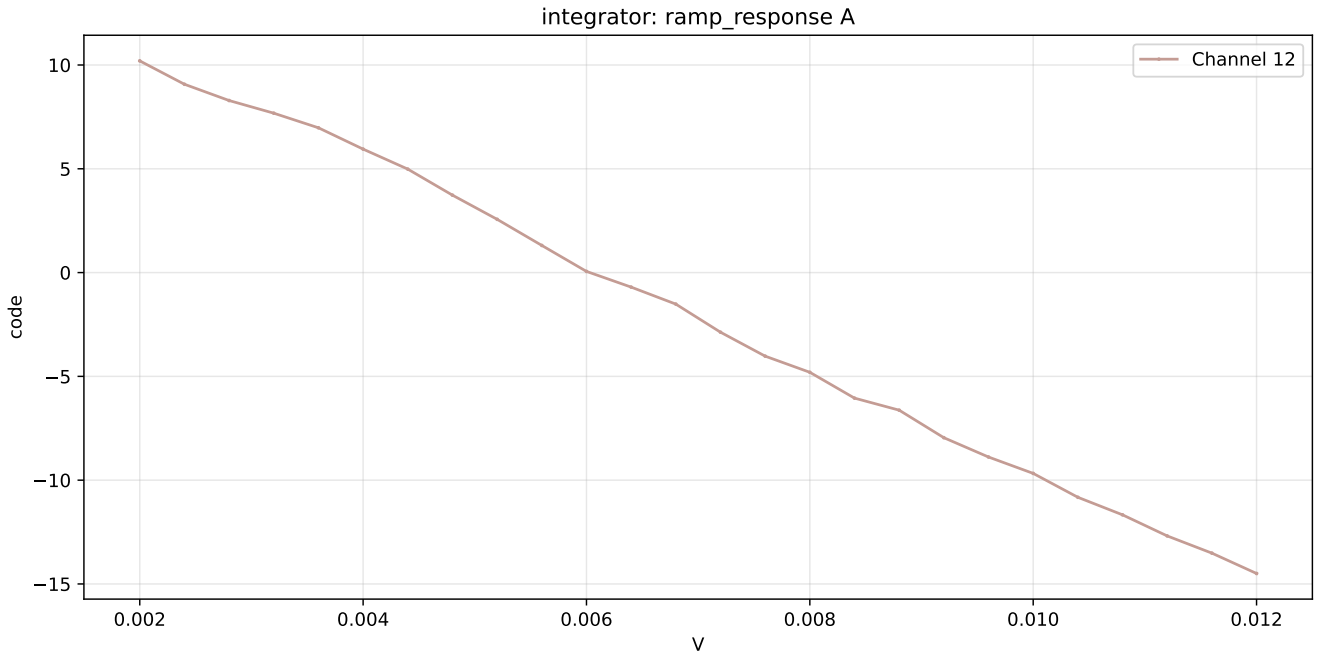


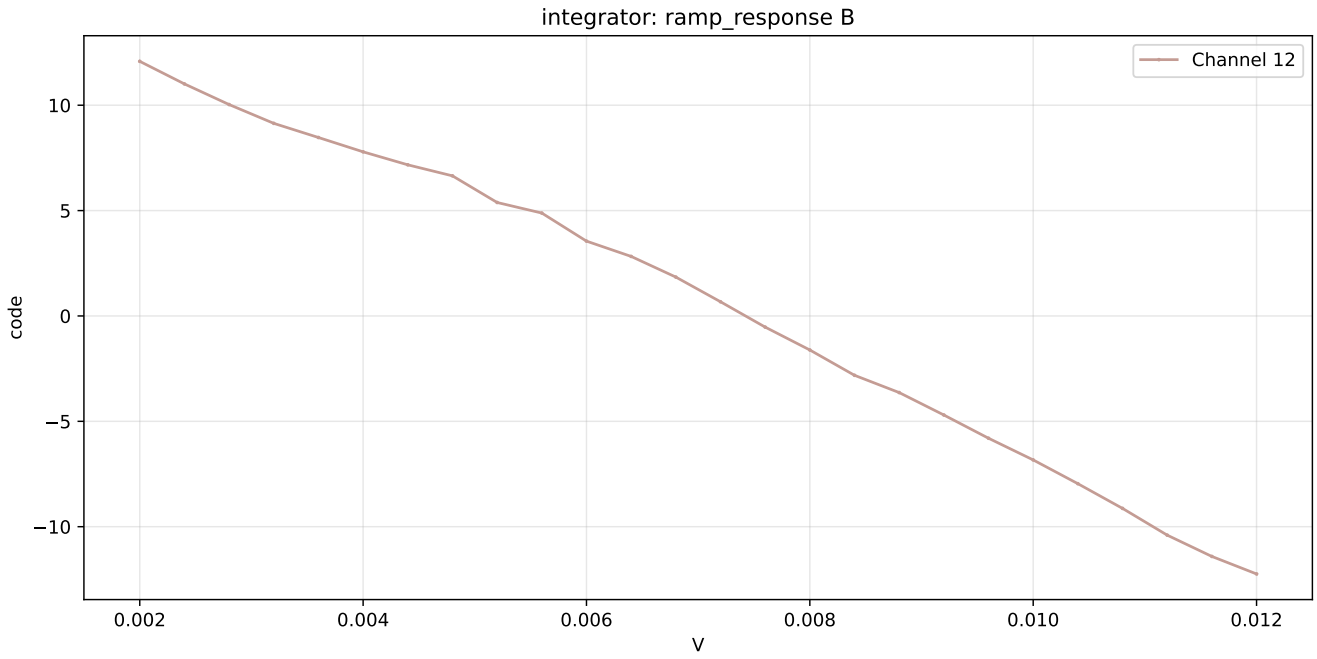

adc-dc: zero_input_time A
code
2220
2200
2180
2160
2140
2120
2100
2080
0
100
200
300
400
500
ns (local capture origin)
Channel 12

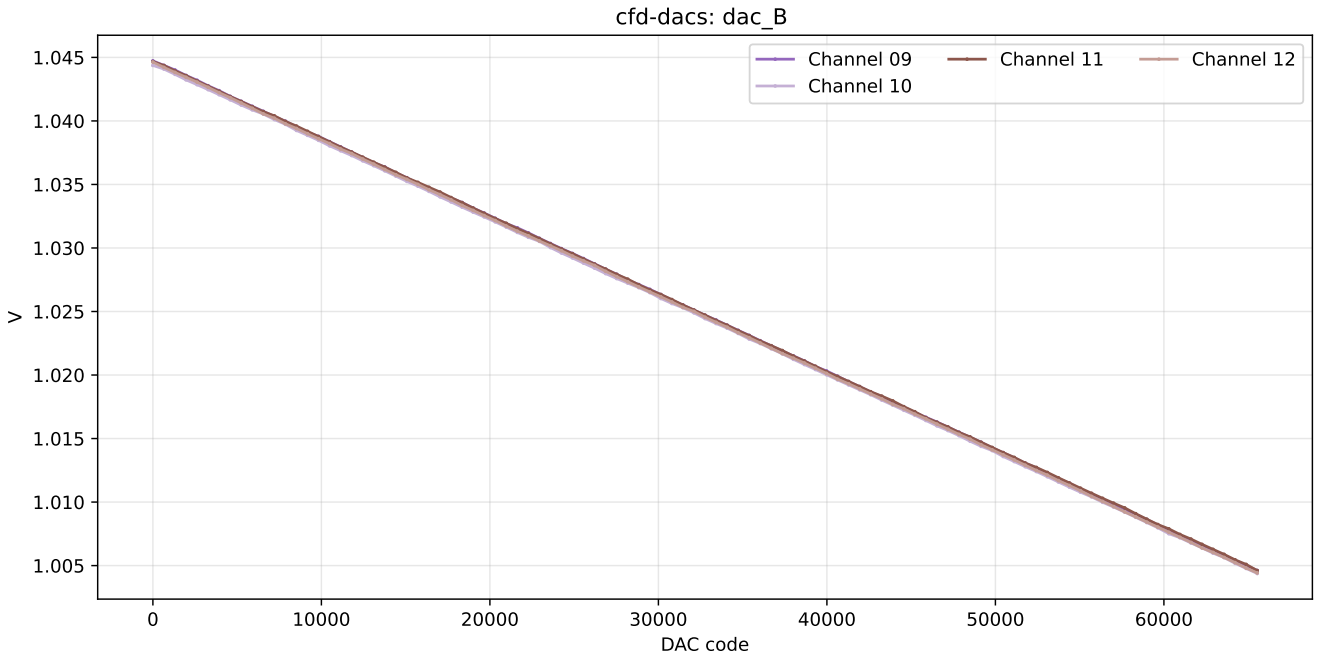

adc-dc: adc_response B
code
4000
3500
3000
2500
2000
1500
1000
500
0
−1.0
−0.5
0.0
0.5
1.0
V
Channel 12

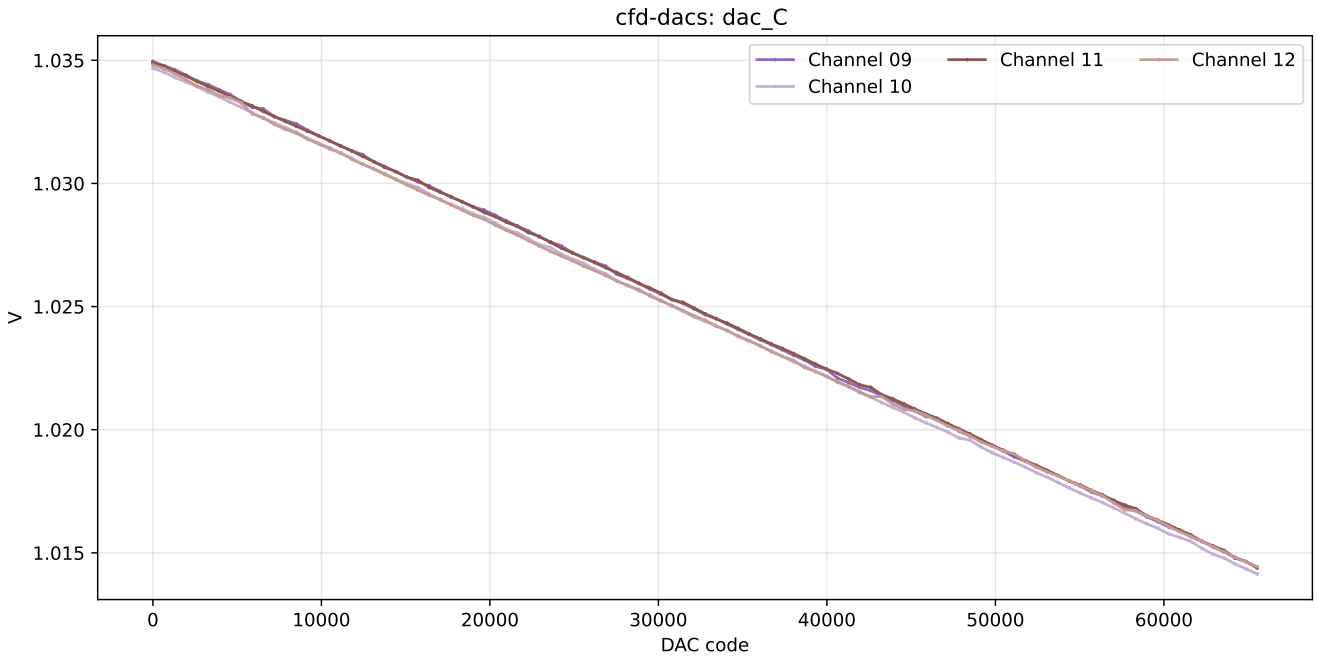

adc-dc: zero_input_fft B
Channel 12
code RMS/bin
0
10
20
30
40
50
60
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Hz (assumed sample rate)
1e7

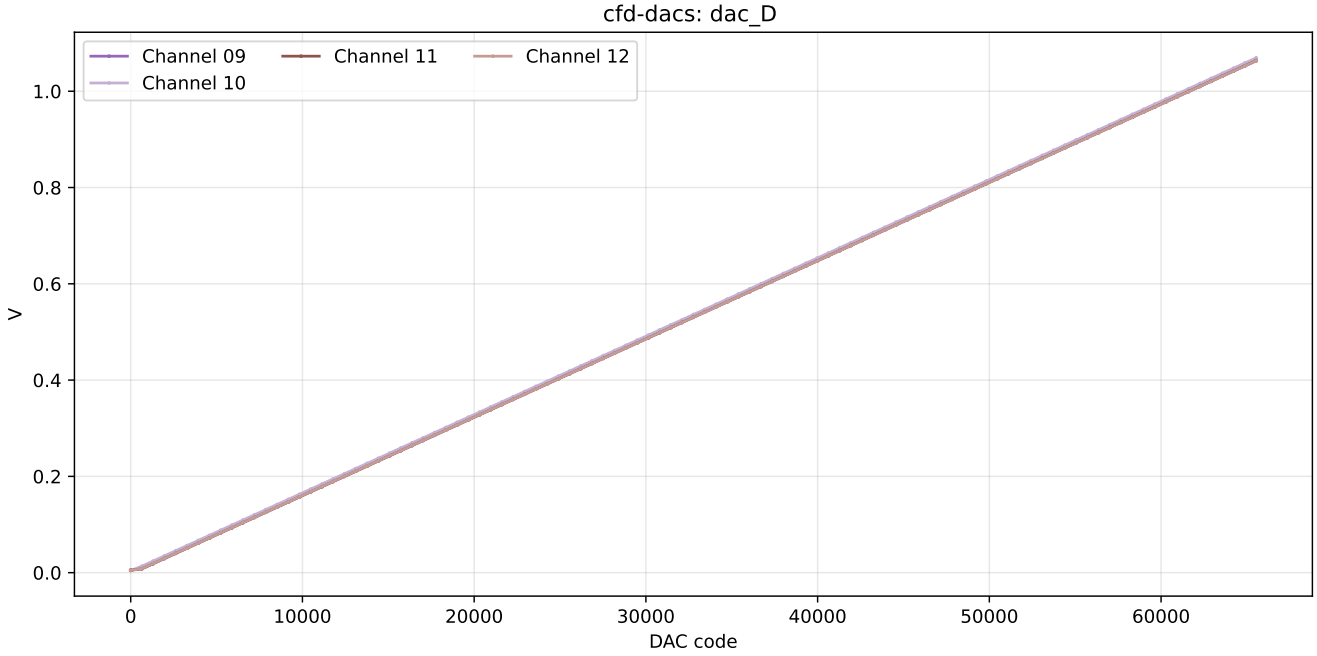

adc-dc: zero_input_time B
Channel 12
code
2075
2100
2125
2150
2175
2200
2225
0
100
200
300
400
500
ns (local capture origin)

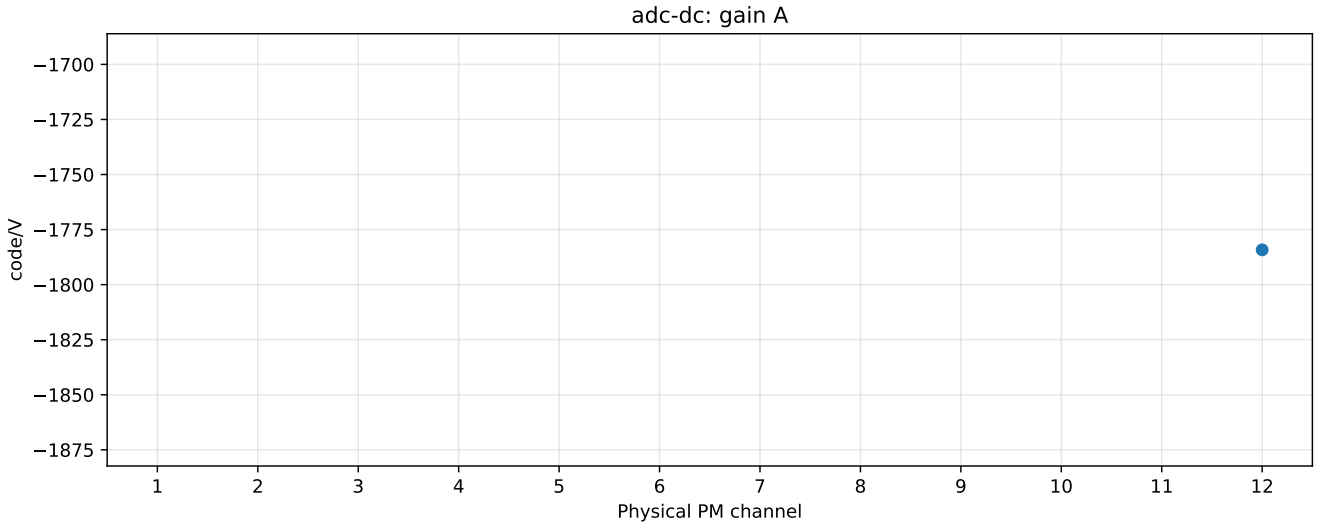

integrator: ramp_response A
Channel 12
code
V
10
5
0
−5
−10
−15
0.002
0.004
0.006
0.008
0.010
0.012

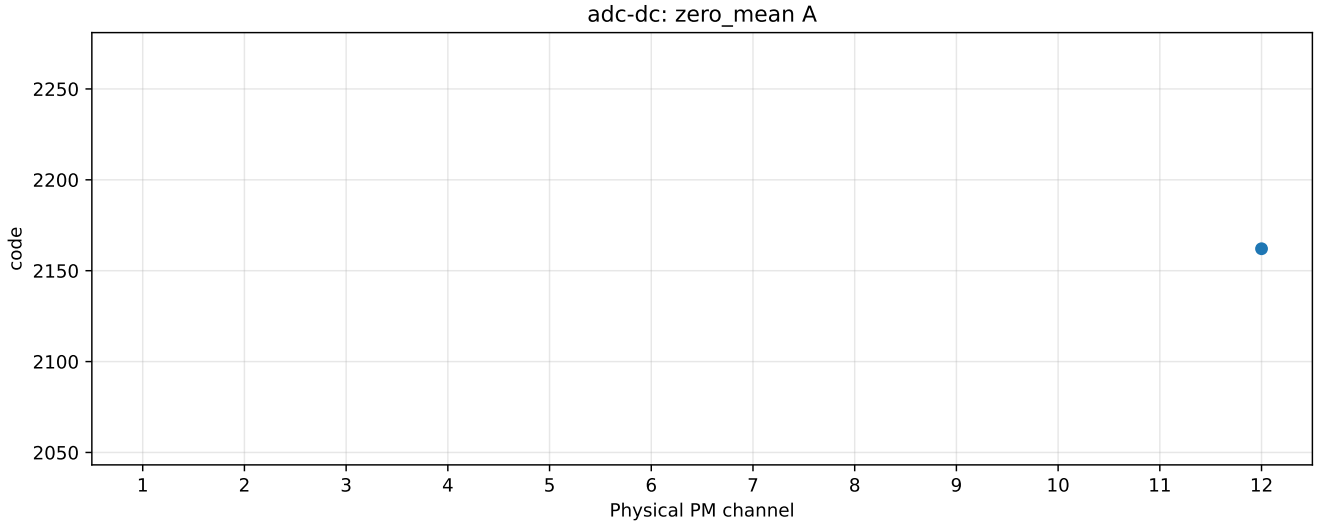

integrator: ramp_response B
Channel 12
code
V
10
5
0
−5
−10
0.002
0.004
0.006
0.008
0.010
0.012

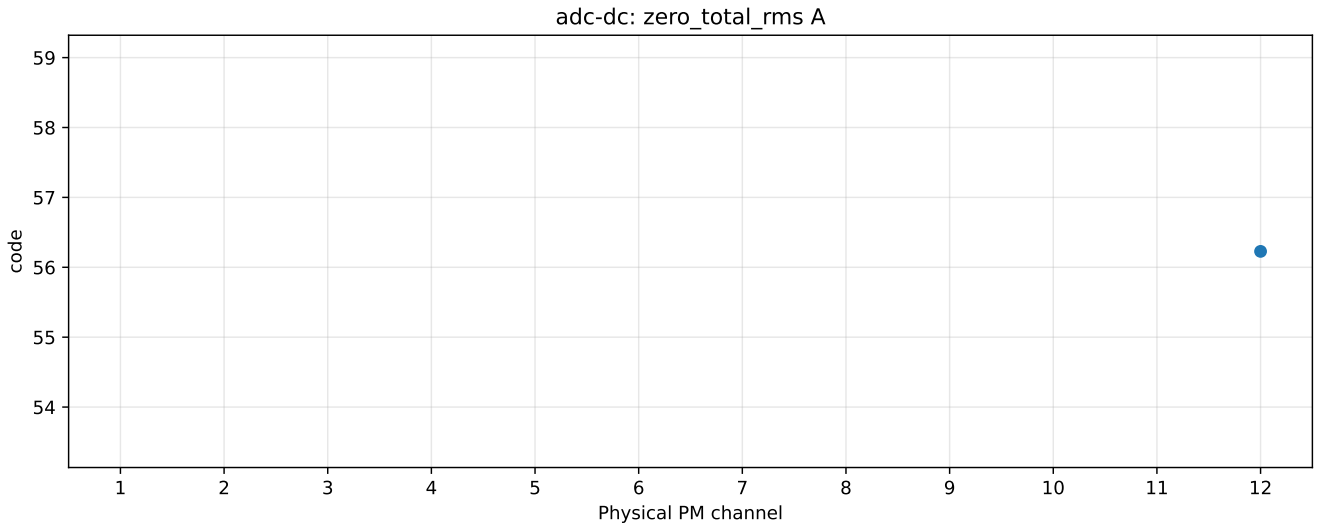

cfd-dacs: dac_B
Channel 09
Channel 10
Channel 11
Channel 12
1.045
1.040
1.035
1.030
1.025
1.020
1.015
1.010
1.005
V
0
10000
20000
30000
40000
50000
60000
DAC code

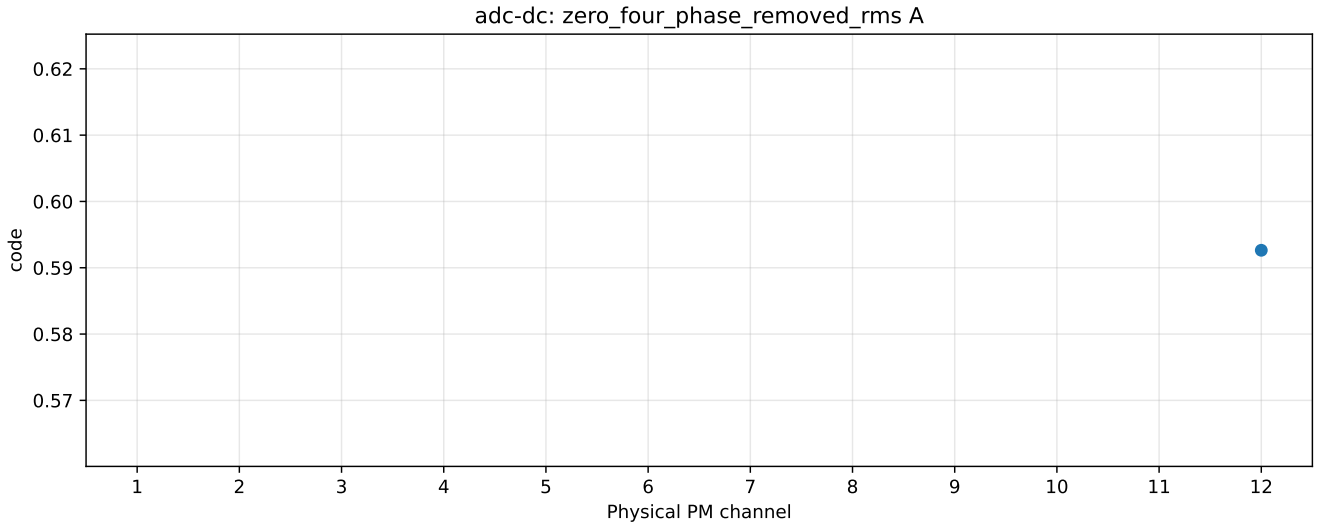

cfd-dacs: dac_C
Channel 09
Channel 10
Channel 11
Channel 12
1.035
1.030
1.025
1.020
1.015
V
0
10000
20000
30000
40000
50000
60000
DAC code

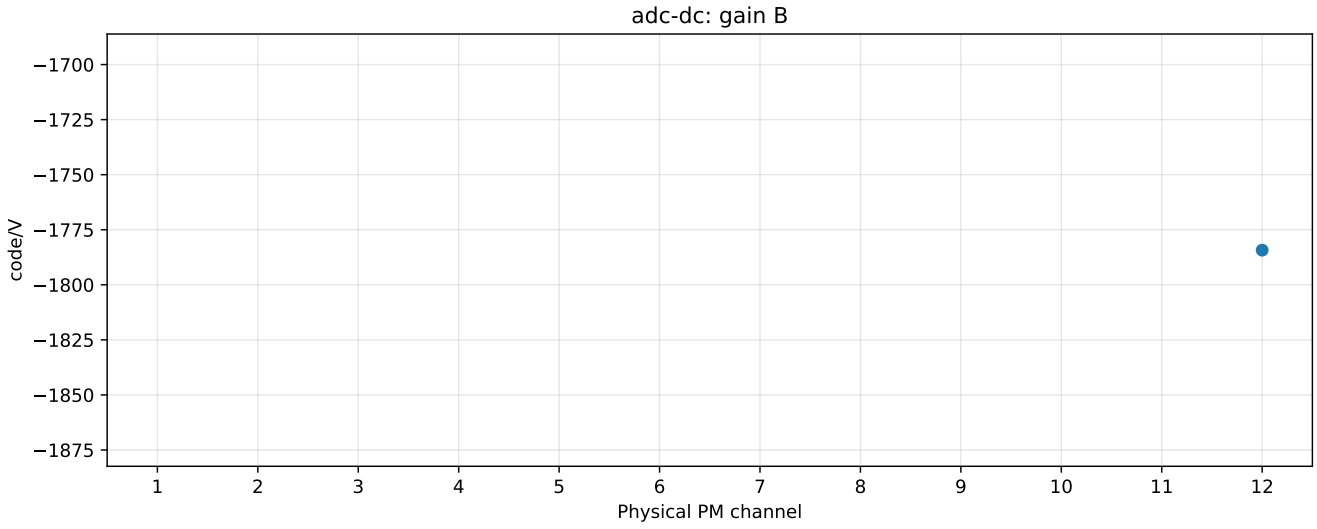

cfd-dacs: dac_D
Channel 09
Channel 10
Channel 11
Channel 12
V
DAC code

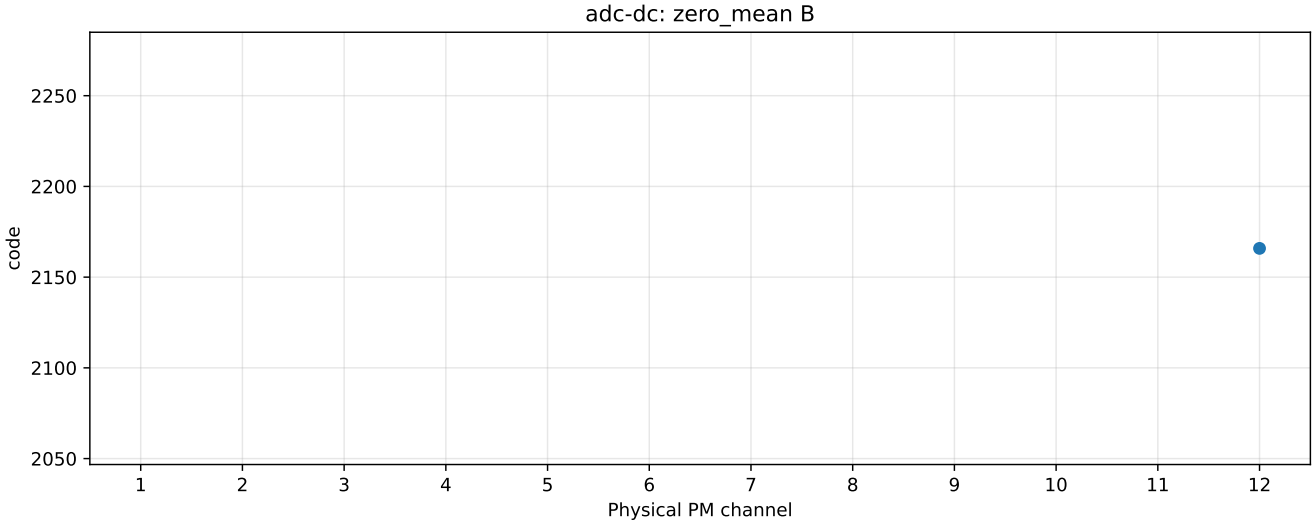

adc-dc: gain A
code/V
Physical PM channel

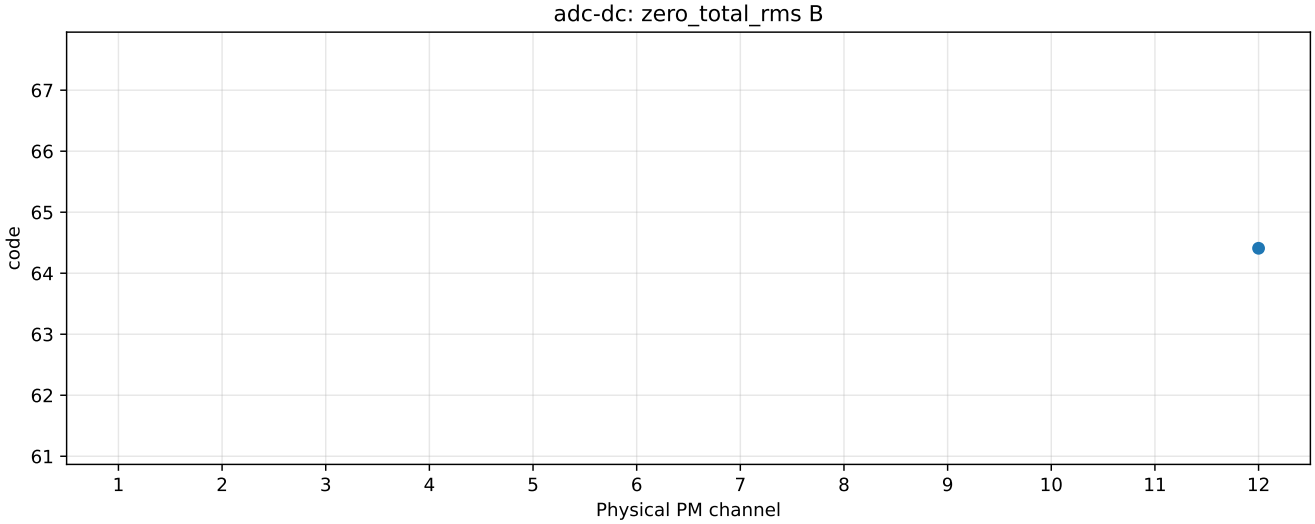

adc-dc: zero_mean A
code
Physical PM channel

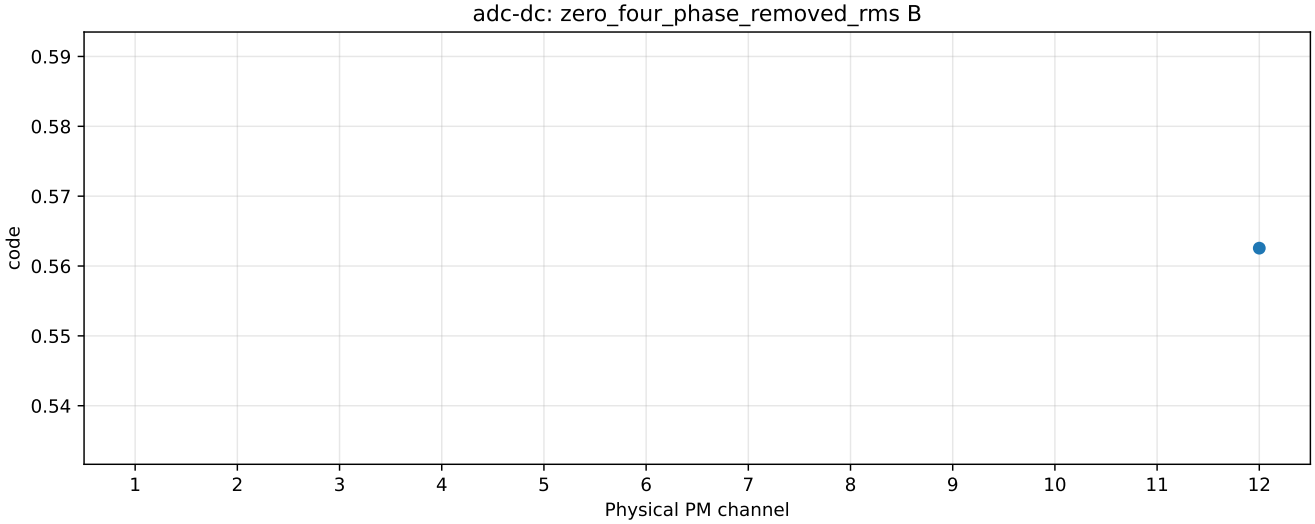

adc-dc: zero_total_rms A
code
59
58
57
56
55
54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

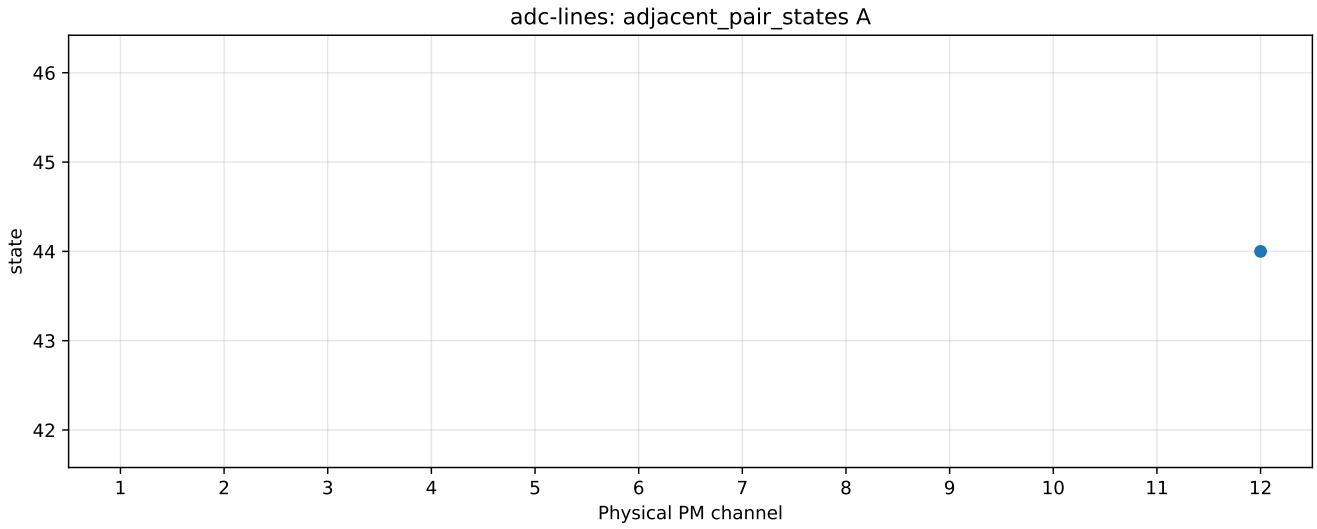

adc-dc: zero_four_phase_removed_rms A
code
0.62
0.61
0.60
0.59
0.58
0.57
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

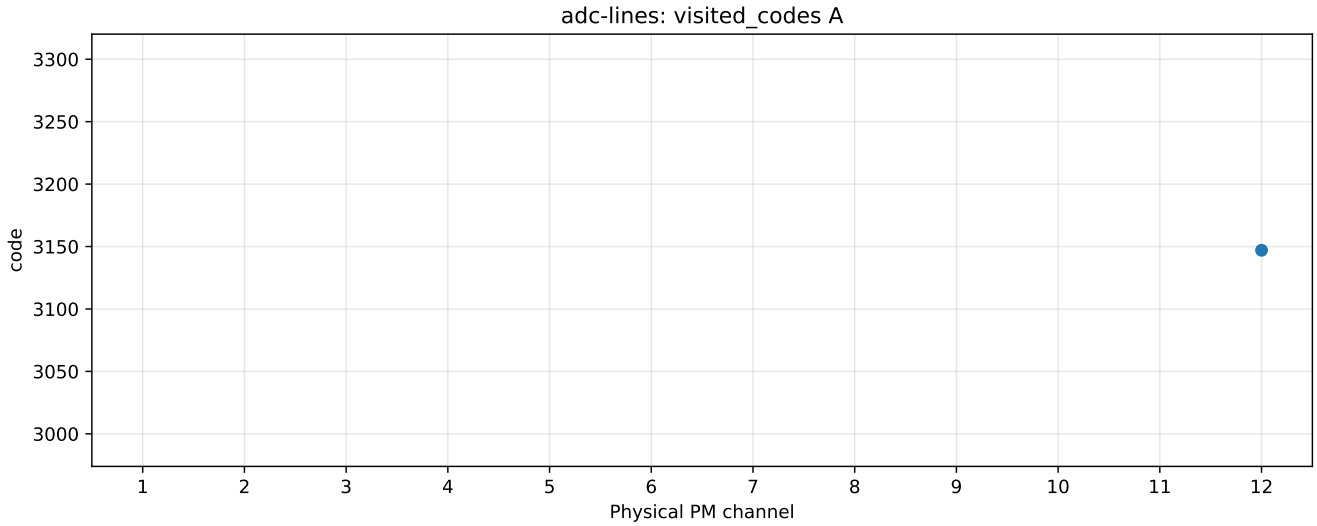

adc-dc: gain B
code/V
−1700
−1725
−1750
−1775
−1800
−1825
−1850
−1875
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

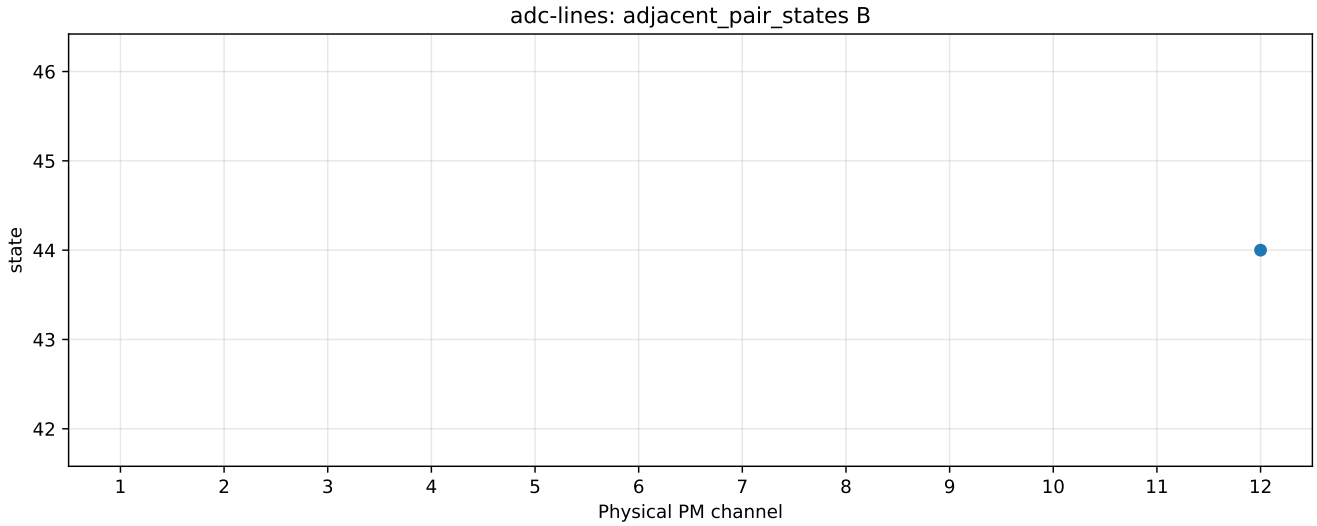

adc-dc: zero_mean B
code
2250
2200
2150
2100
2050
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

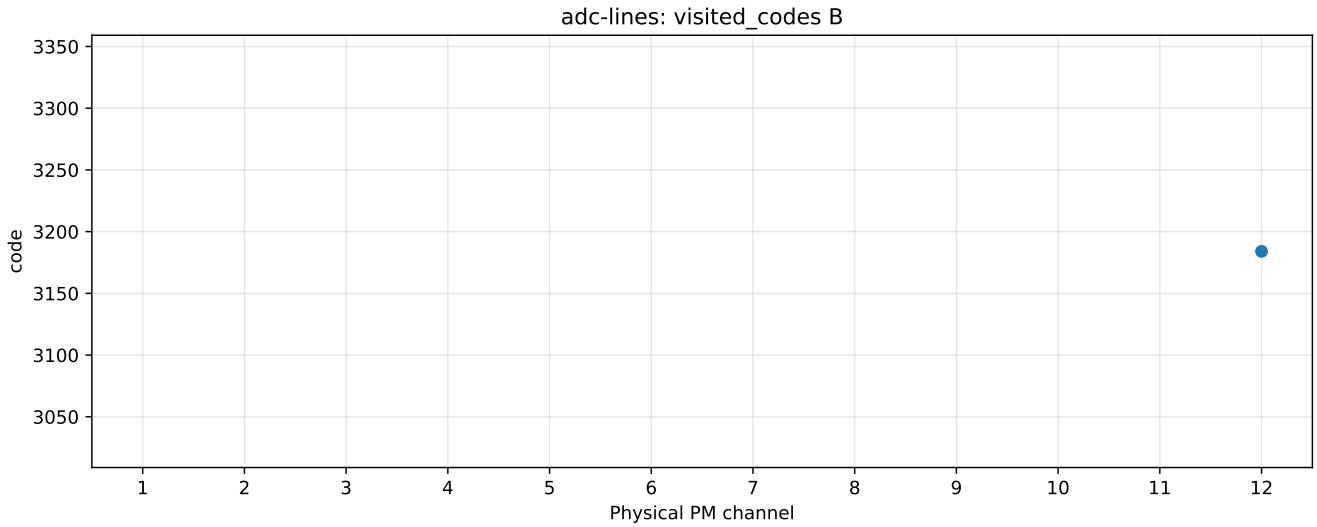

adc-dc: zero_total_rms B
code
67
66
65
64
63
62
61
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

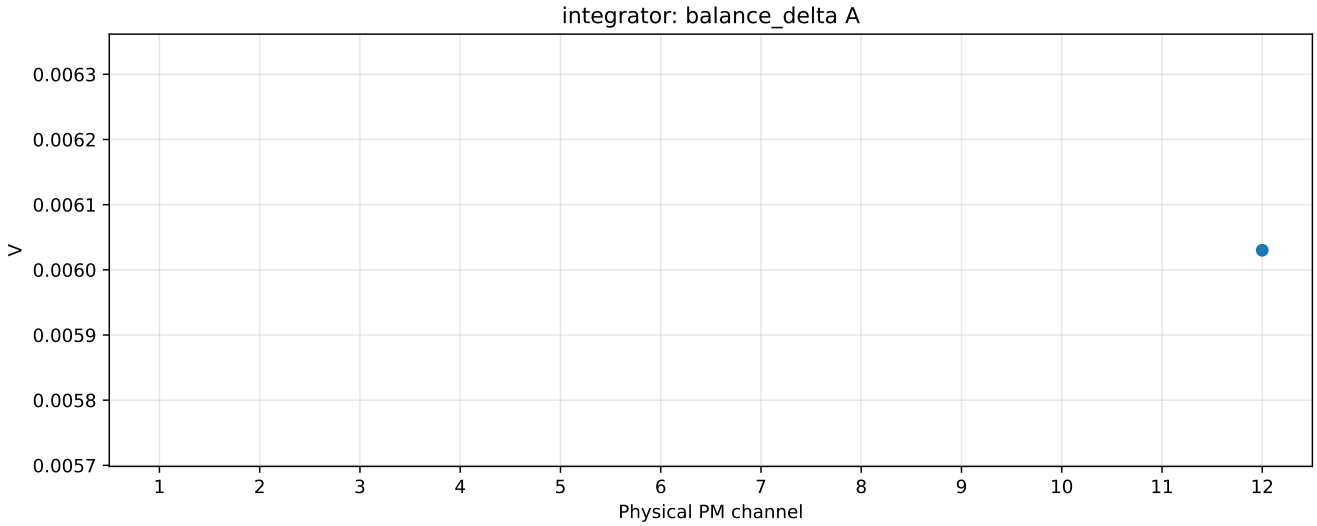

adc-dc: zero_four_phase_removed_rms B
code
0.59
0.58
0.57
0.56
0.55
0.54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

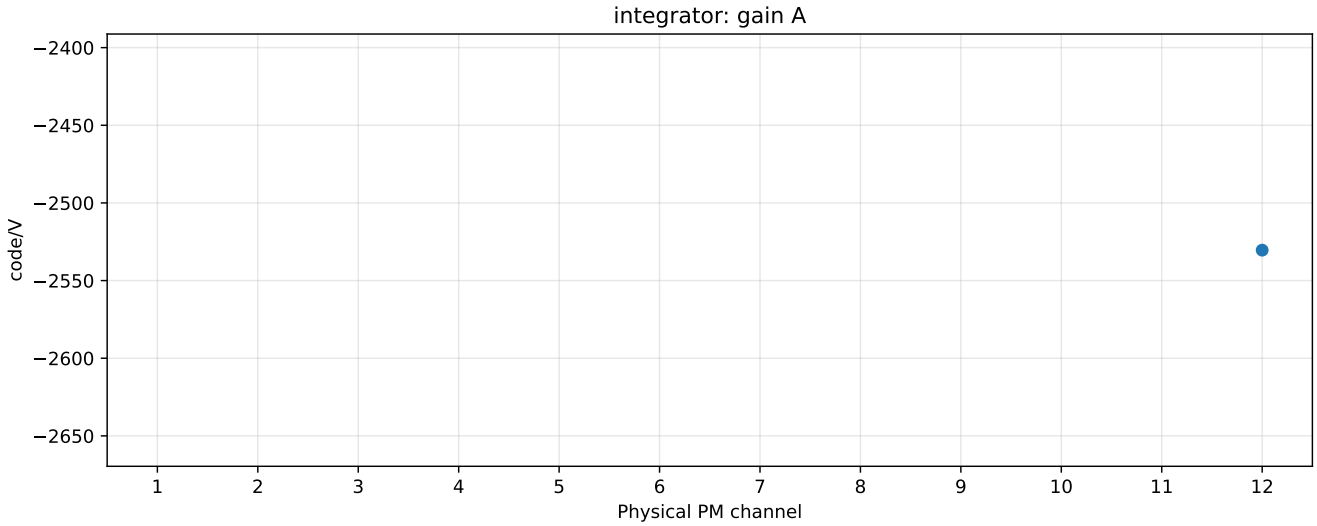

adc-lines: adjacent_pair_states A
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

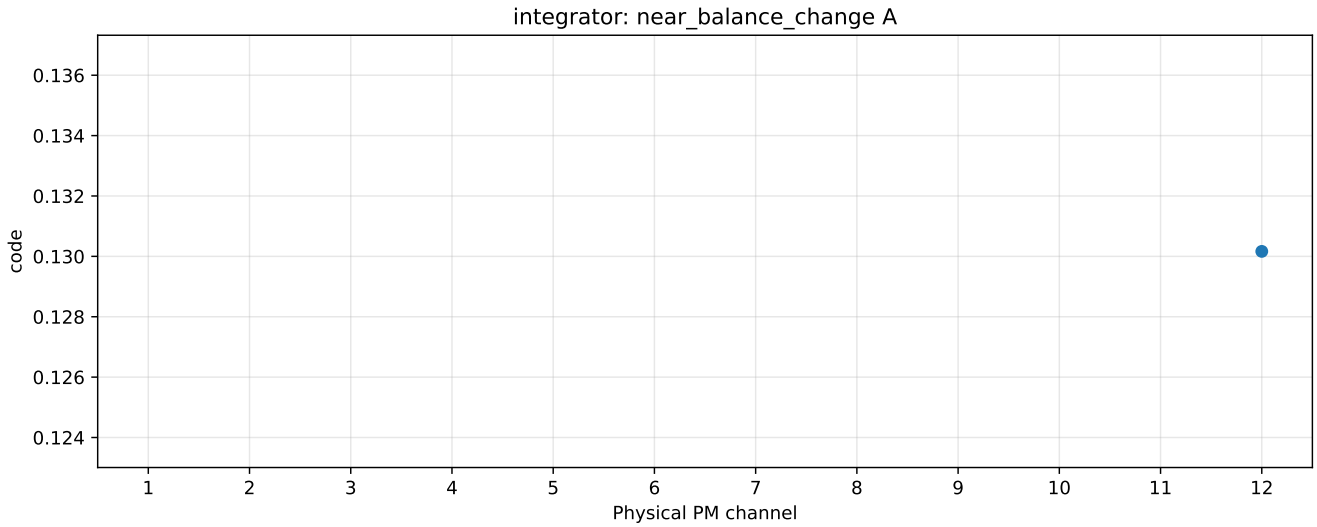

adc-lines: visited_codes A
code
3300
3250
3200
3150
3100
3050
3000
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

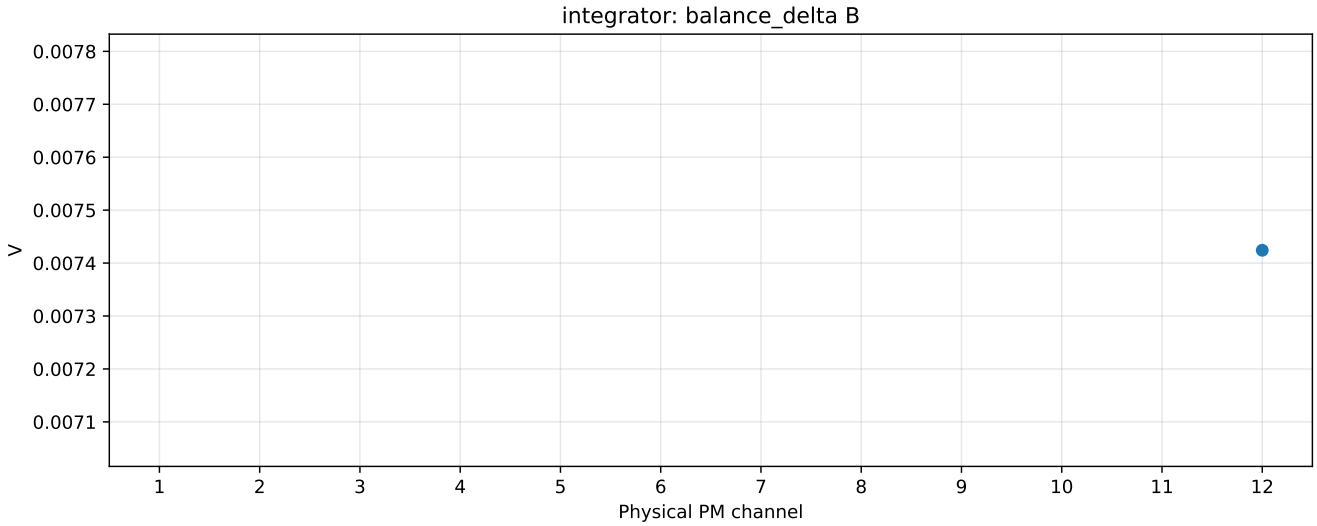

adc-lines: adjacent_pair_states B
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

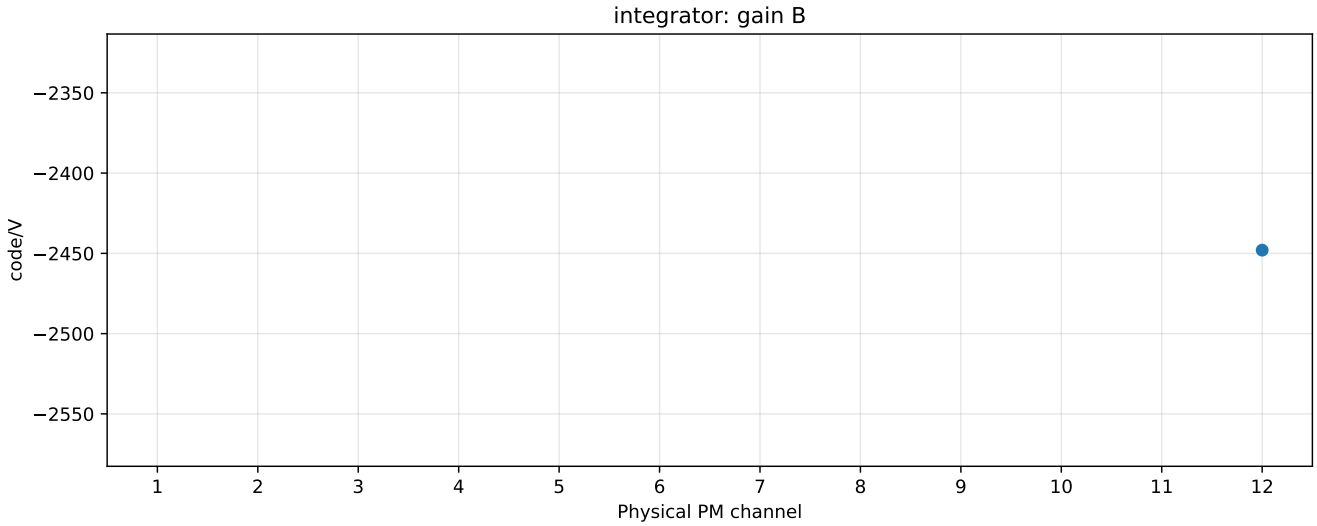

adc-lines: visited_codes B
code
3350
3300
3250
3200
3150
3100
3050
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

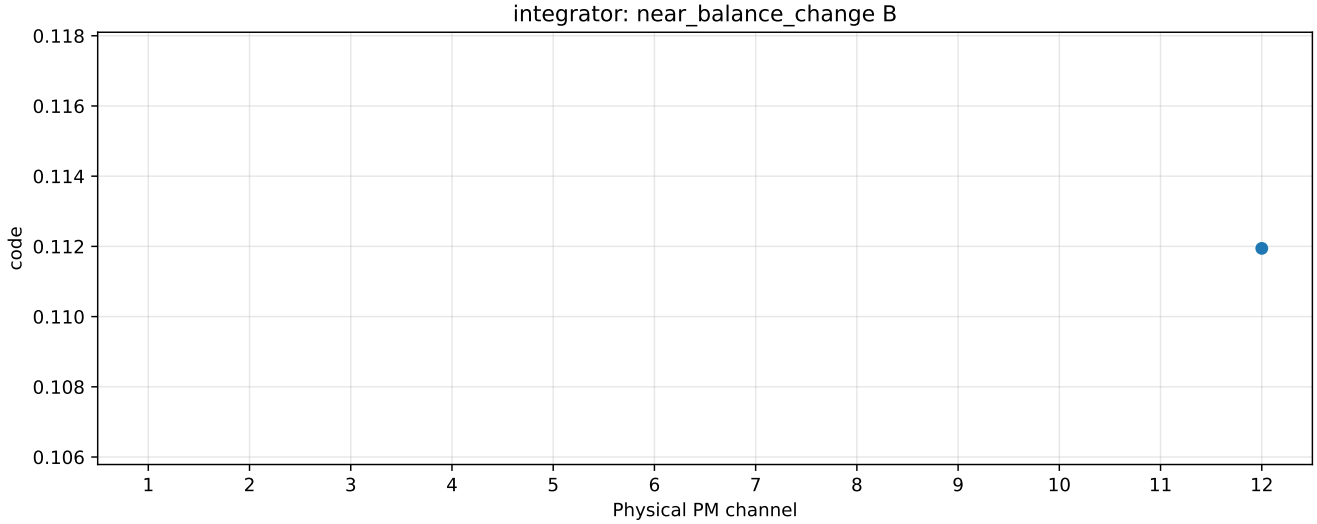

integrator: balance_delta A
V
0.0063
0.0062
0.0061
0.0060
0.0059
0.0058
0.0057
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

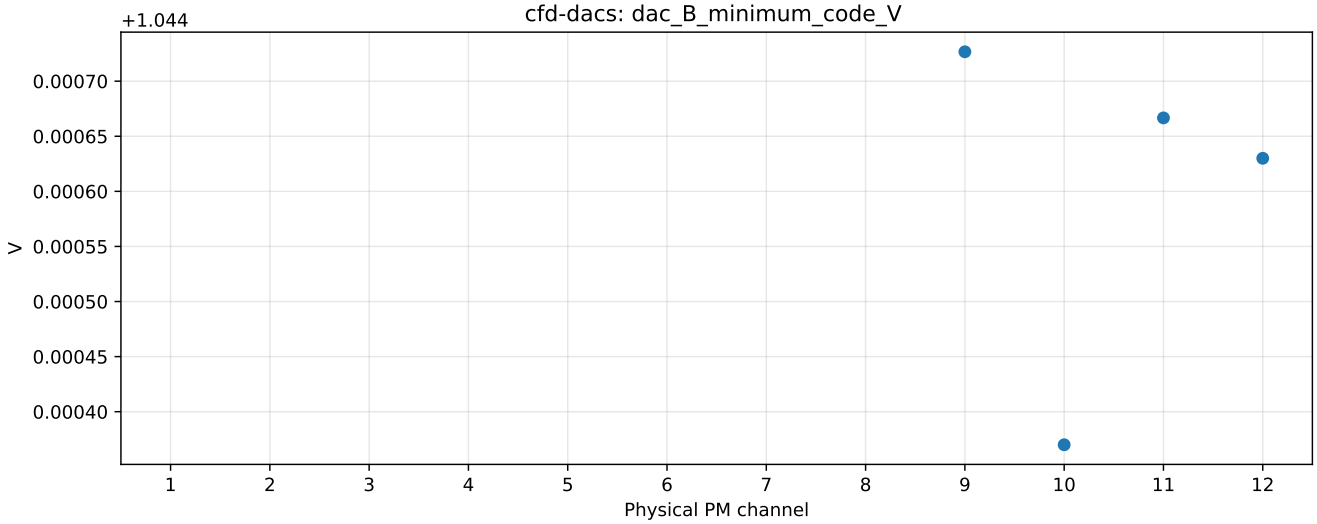

integrator: gain A
code/V
−2400
−2450
−2500
−2550
−2600
−2650
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

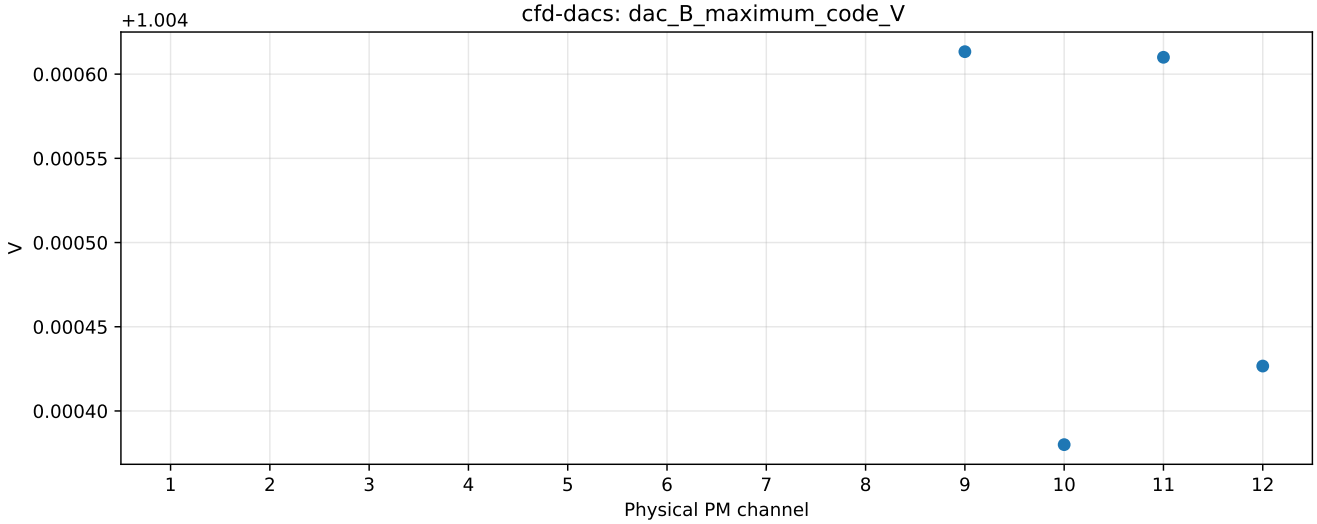

integrator: near_balance_change A
code
0.136
0.134
0.132
0.130
0.128
0.126
0.124
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

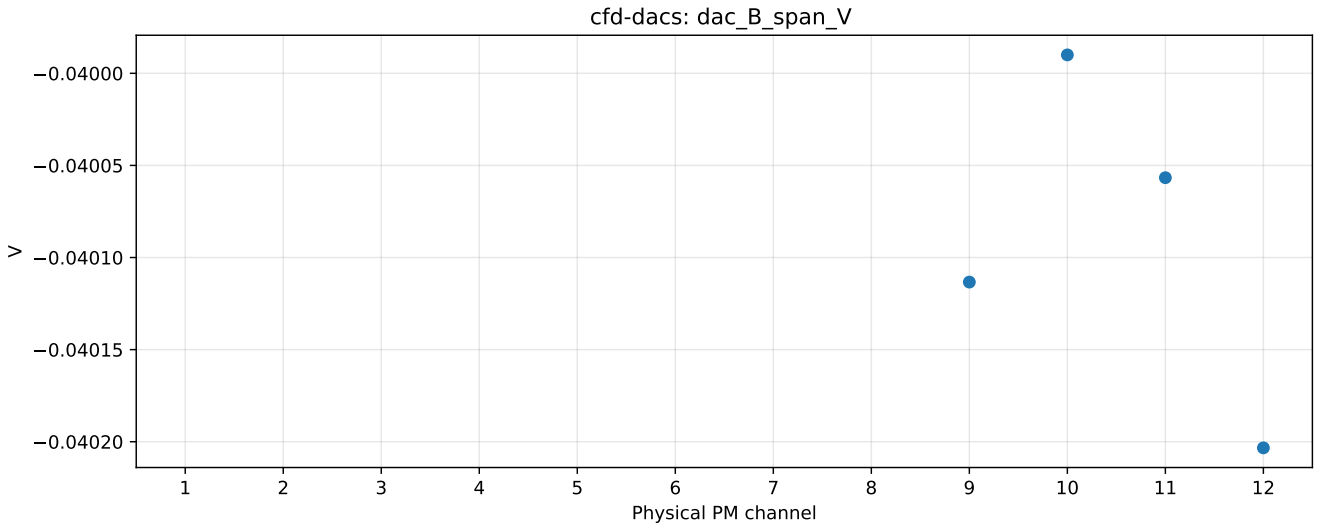

integrator: balance_delta B
V
0.0078
0.0077
0.0076
0.0075
0.0074
0.0073
0.0072
0.0071
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

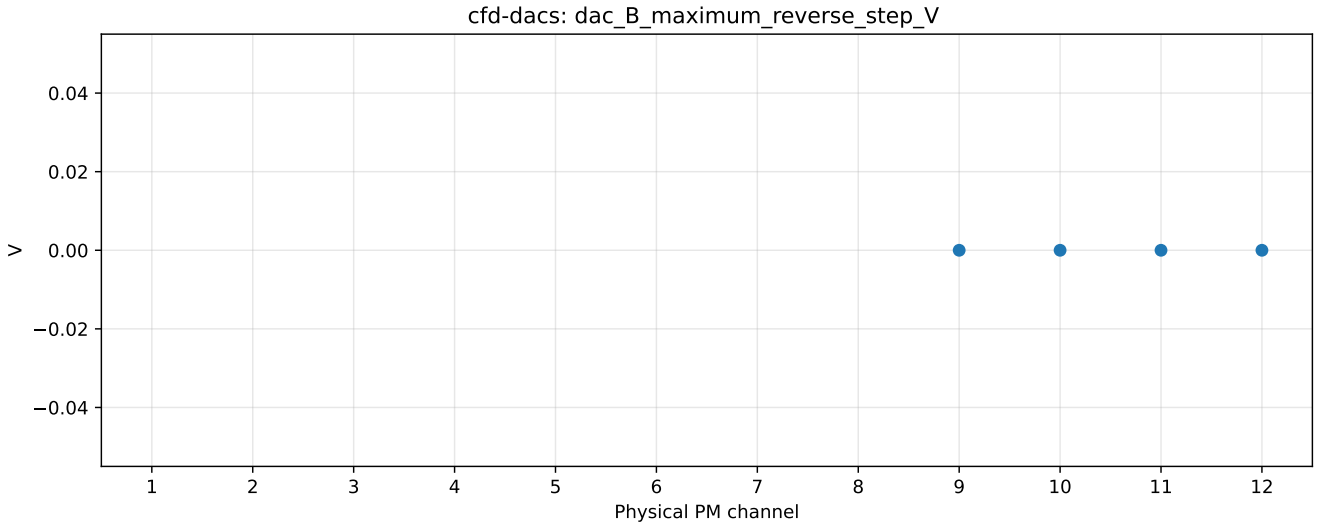

integrator: gain B
code/V
−2350
−2400
−2450
−2500
−2550
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

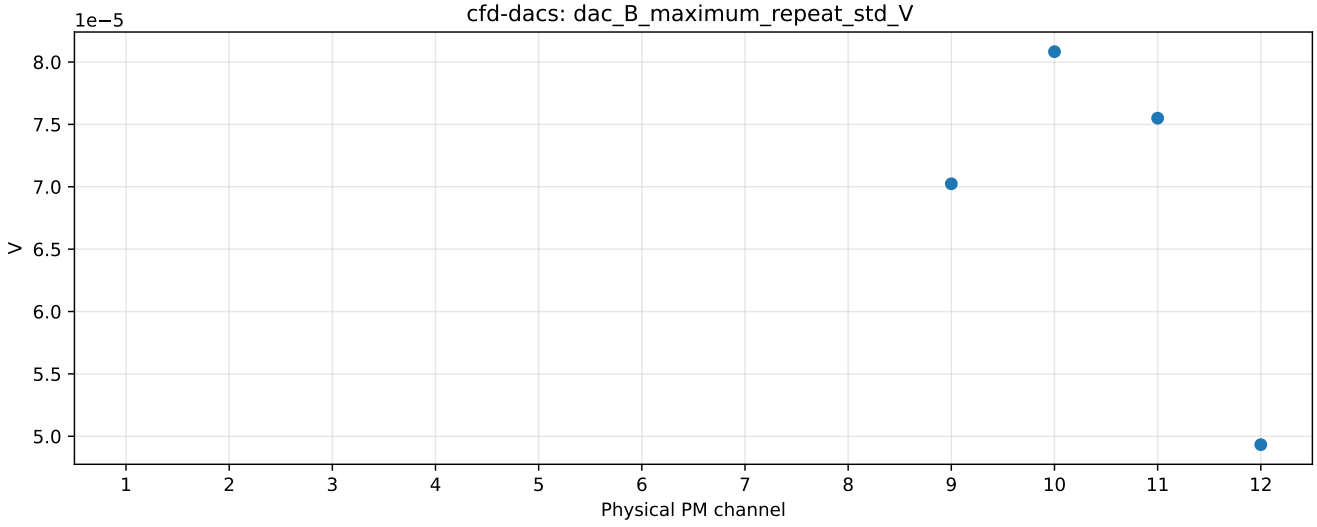

integrator: near_balance_change B
code
0.118
0.116
0.114
0.112
0.110
0.108
0.106
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

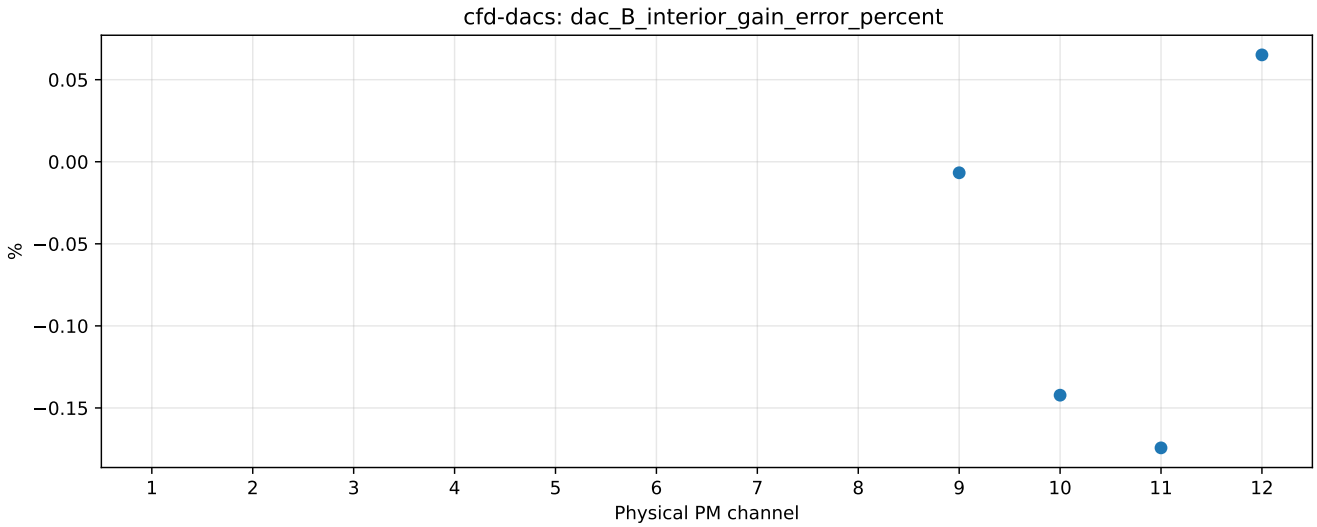

cfd-dacs: dac_B_minimum_code_V
+1.044
V
0.00070
0.00065
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

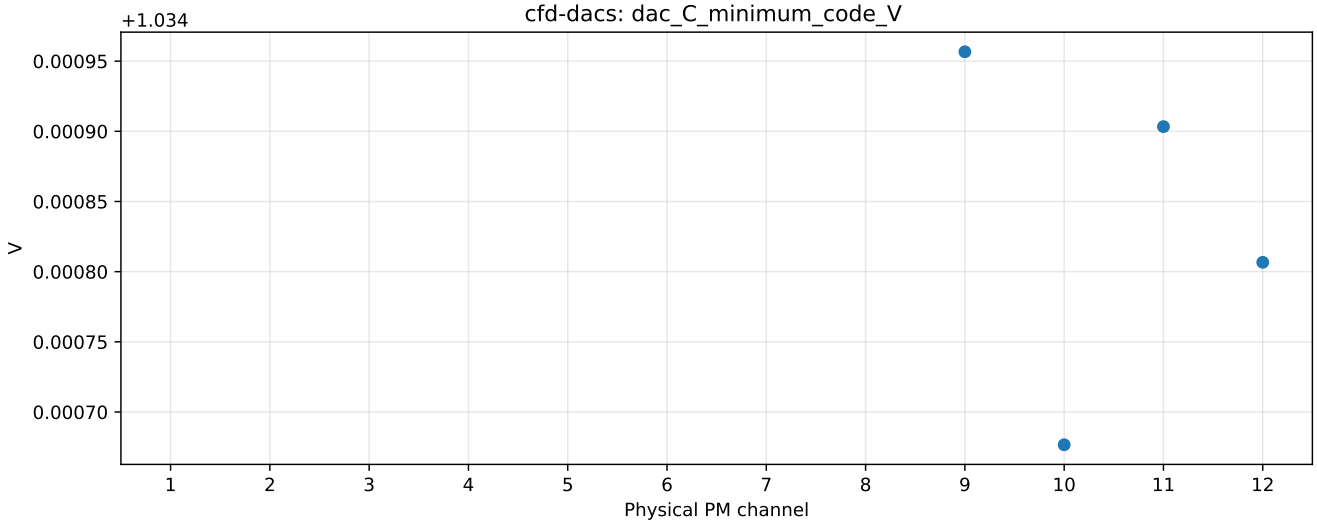

cfd-dacs: dac_B_maximum_code_V
+1.004
V
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

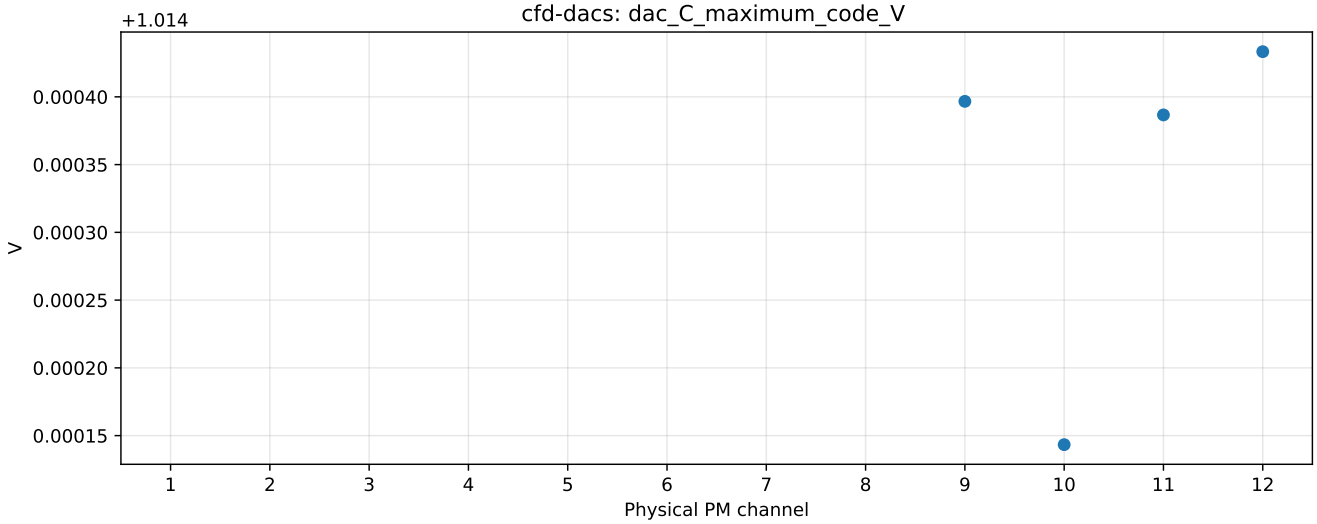

cfd-dacs: dac_B_span_V
−0.04000
−0.04005
−0.04010
−0.04015
−0.04020
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

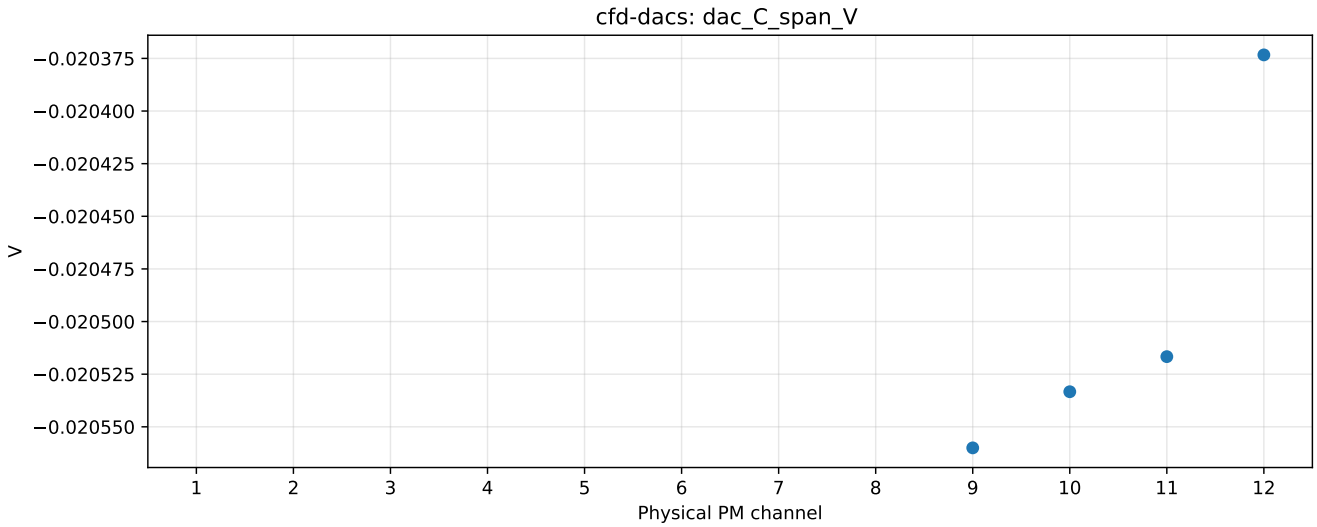

cfd-dacs: dac_B_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

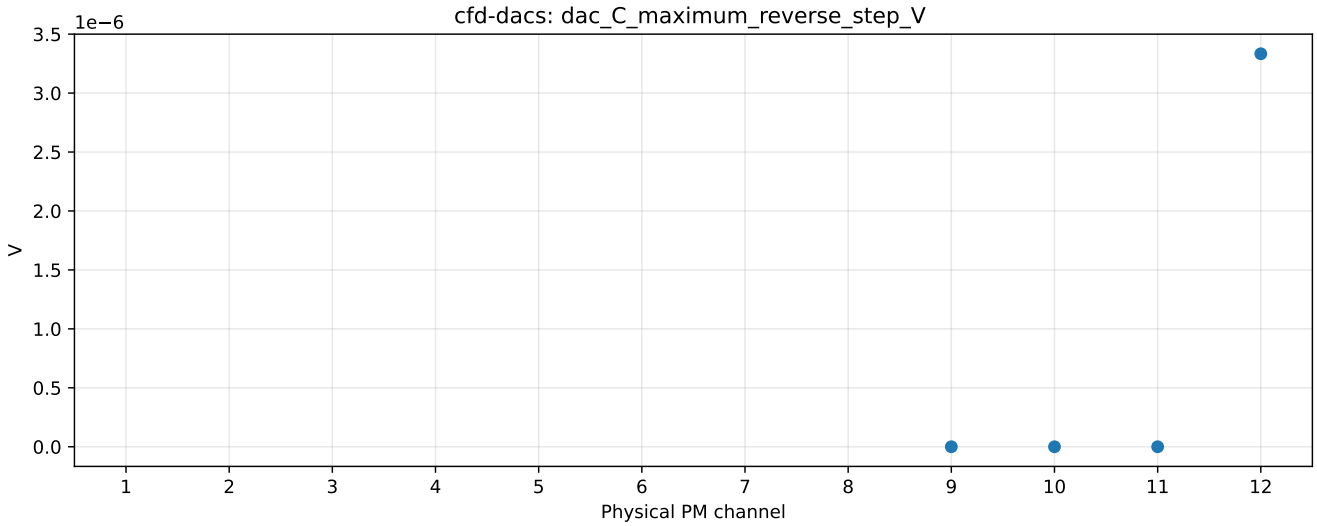

cfd-dacs: dac_B_maximum_repeat_std_V
1e−5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

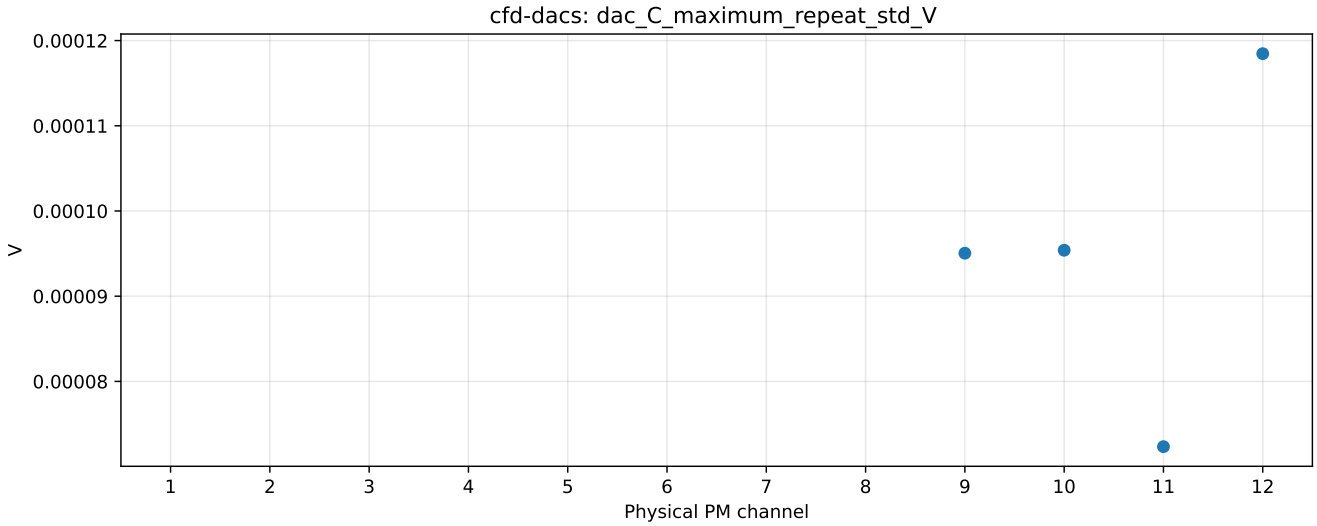

cfd-dacs: dac_B_interior_gain_error_percent
0.05
0.00
−0.05
−0.10
−0.15
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

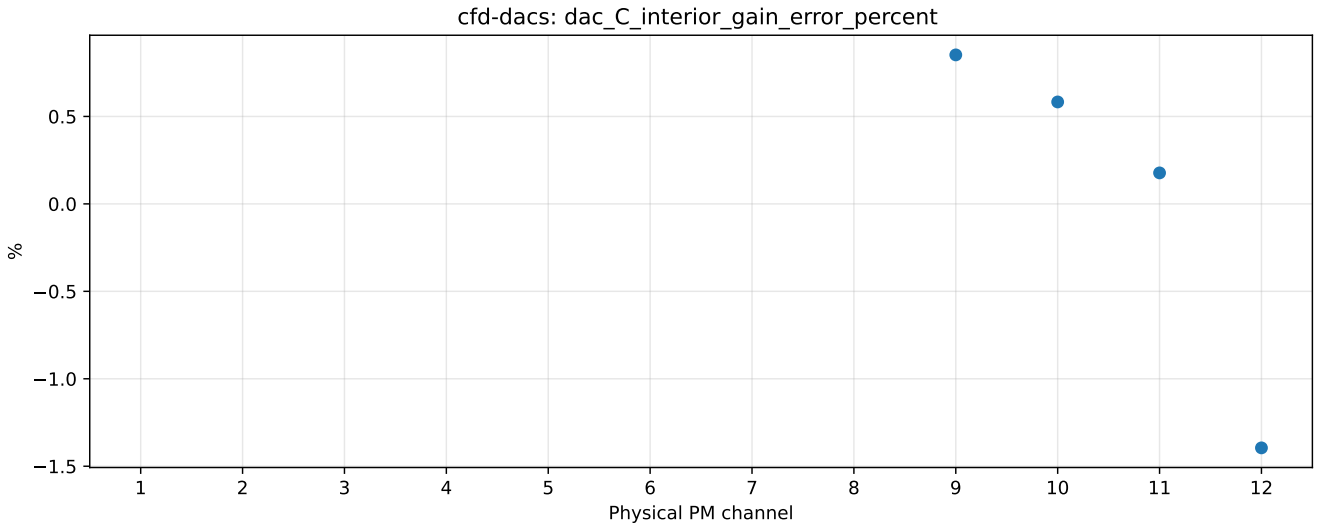

cfd-dacs: dac_C_minimum_code_V
+1.034
0.00095
0.00090
0.00085
0.00080
0.00075
0.00070
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

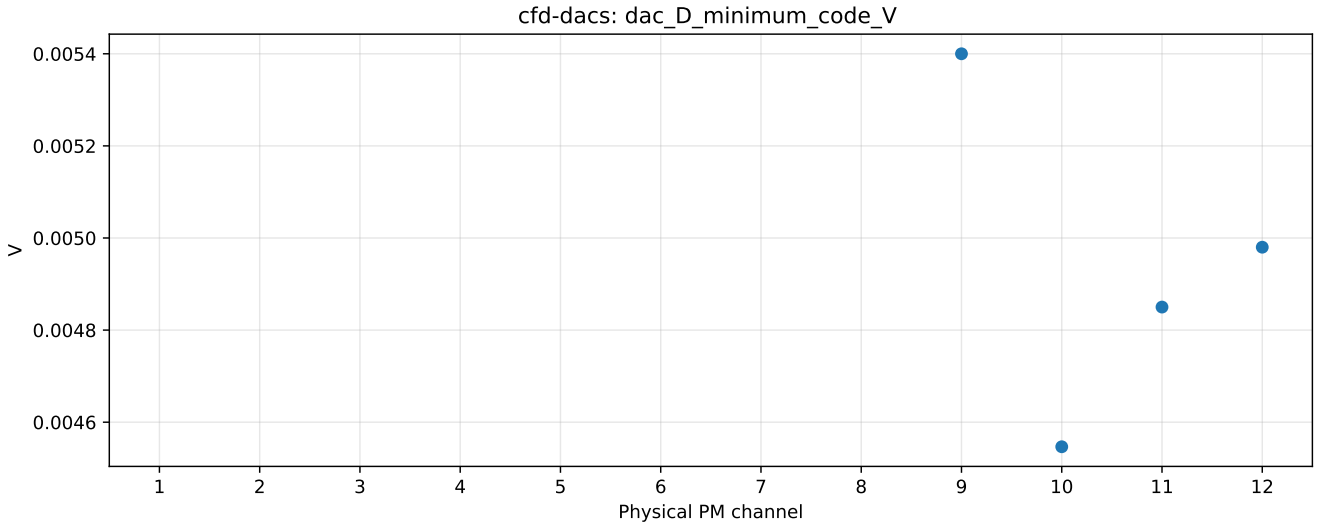

cfd-dacs: dac_C_maximum_code_V
+1.014
0.00040
0.00035
0.00030
0.00025
0.00020
0.00015
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

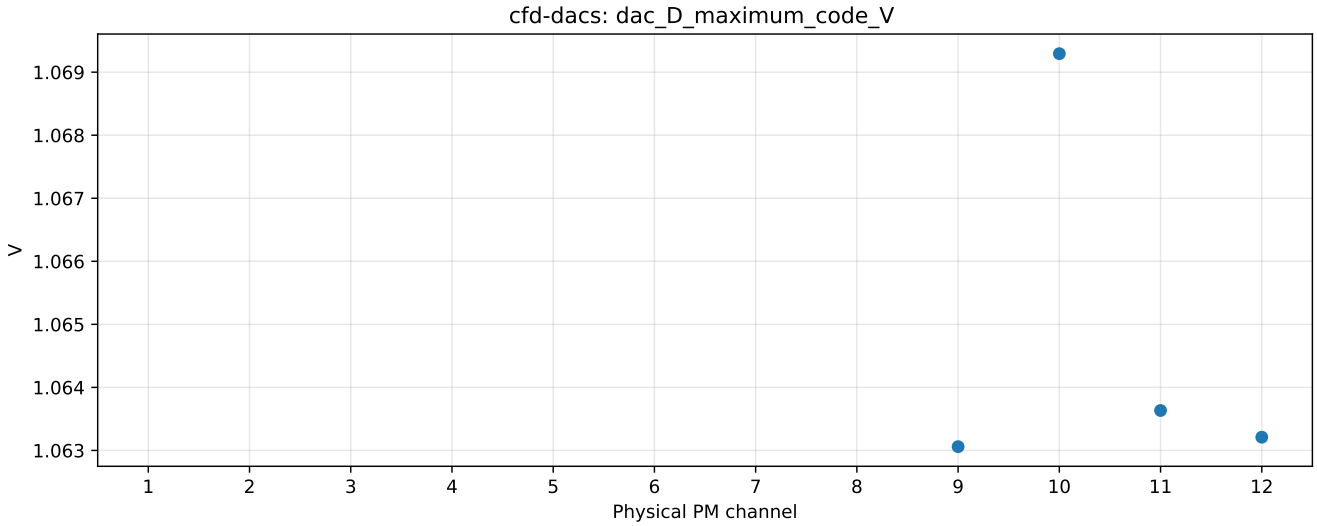

cfd-dacs: dac_C_span_V
−0.020375
−0.020400
−0.020425
−0.020450
−0.020475
−0.020500
−0.020525
−0.020550
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

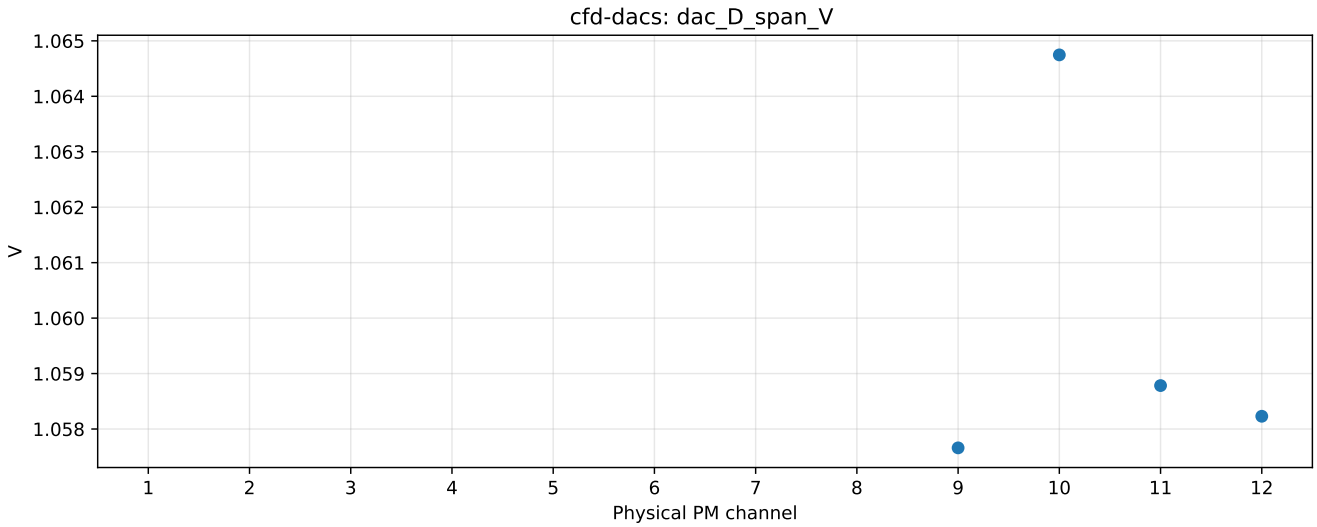

cfd-dacs: dac_C_maximum_reverse_step_V
1e−6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

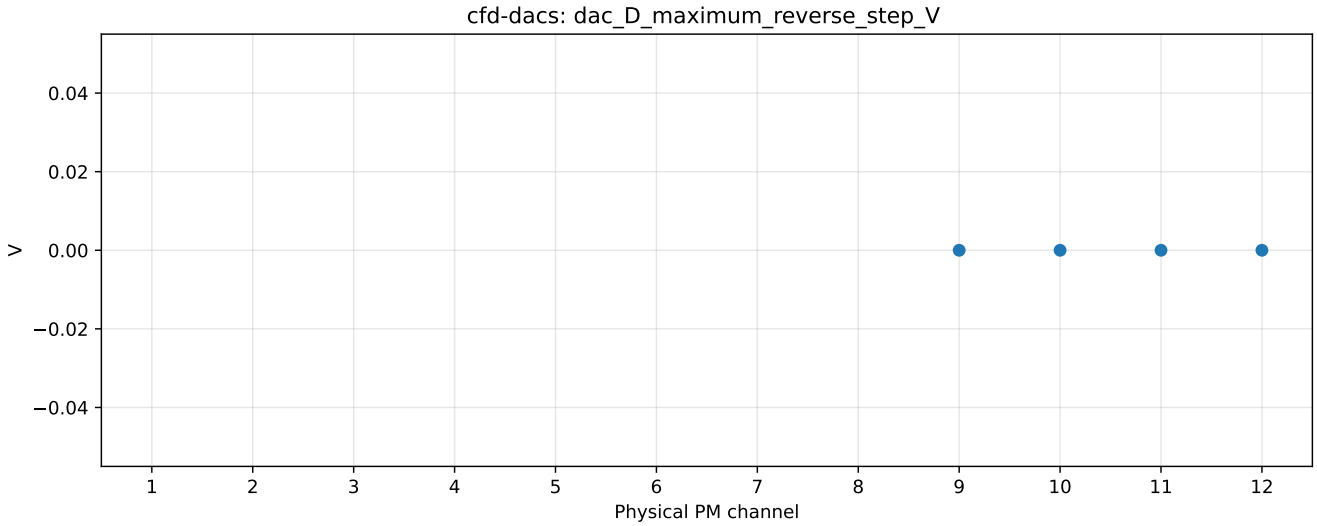

cfd-dacs: dac_C_maximum_repeat_std_V
0.00012
0.00011
0.00010
0.00009
0.00008
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

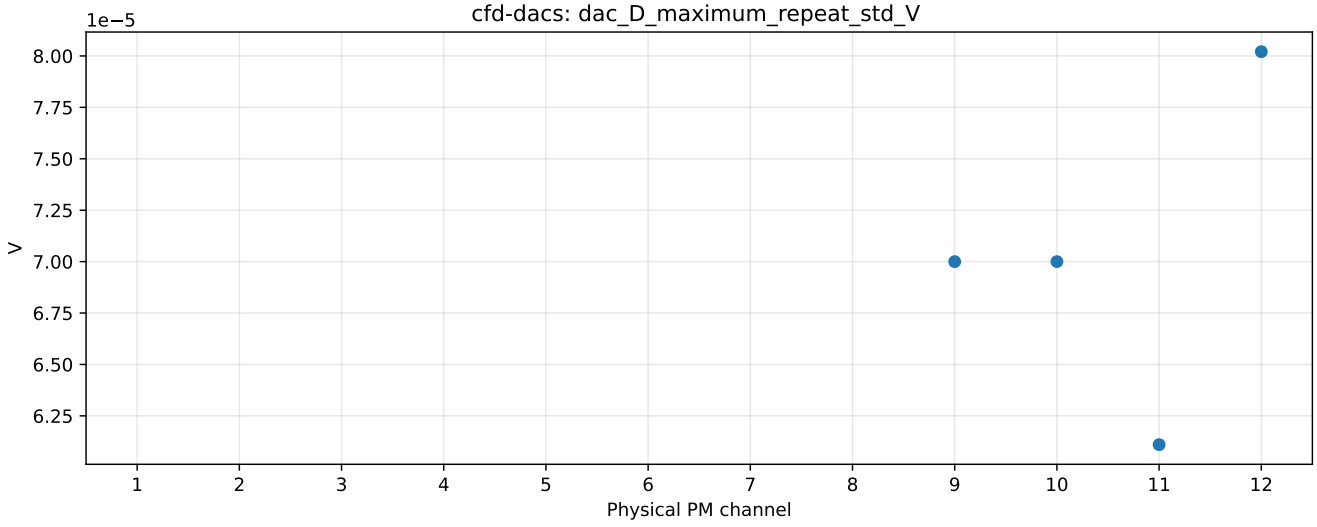

cfd-dacs: dac_C_interior_gain_error_percent
%
0.5
0.0
−0.5
−1.0
−1.5
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

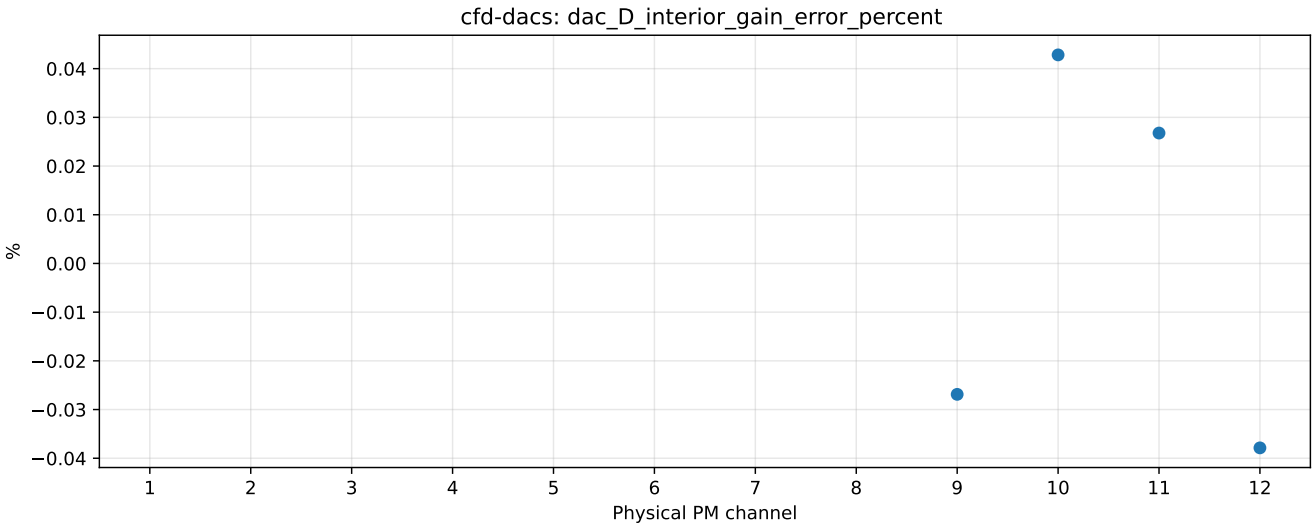

cfd-dacs: dac_D_minimum_code_V
V
0.0054
0.0052
0.0050
0.0048
0.0046
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-dacs: dac_D_maximum_code_V
V
1.069
1.068
1.067
1.066
1.065
1.064
1.063
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-dacs: dac_D_span_V
1.065
1.064
1.063
1.062
1.061
1.060
1.059
1.058
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-dacs: dac_D_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-dacs: dac_D_maximum_repeat_std_V
1e−5
8.00
7.75
7.50
7.25
7.00
6.75
6.50
6.25
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-dacs: dac_D_interior_gain_error_percent
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel